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Fusion by beam ions in a low collisionality, high mirror ratio magnetic mirror

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Abstract

The classical problem of neutral beam ions slowing down in a magnetic mirror geometry is revisited to provide predictive capability for the new Wisconsin HTS Axisymmetric Mirror under development at the University of Wisconsin. A Fokker–Planck model named fast beam ion solver (FBIS) is developed to include the spatial non-uniformity of a physical mirror geometry. The mathematical framework allows for efficient orbit averaging of the pitch-angle scattering operator, and permits a determination of the axial profile of the ambipolar potential confining the electrons. The numerical results from FBIS are consistent with earlier work, but further show how mirror-ratio and a near square magnetic well optimizes the fusion gain. The numerical results are also applied to inform the conceptual design of WHAM++, a low capital-cost breakeven-class magnetic mirror device.

Keywords: neutral beams, magnetic mirror, Fokker–Planck model

(Some figures may appear in colour only in the online journal)

1. Introduction

The first fusion experiment of any kind, magnetic or inertial, to achieve the 10 keV ion temperatures required for DT fusion was the 2XIIB mirror heated by 20 keV neutral beam injection (NBI) [1], ultimately reaching a normalized pressure of $\beta = 2\mu_0 nT/B^2 \sim 1$ [2] and increasing its mirror ratio from ~ 2 to >10 via diamagnetism.

The yin-yang magnet design of 2XIIB achieved MHD stabilization everywhere, leaving only the drift cyclotron loss-cone instability [3]. With hindsight, reorienting the beams to produce so-called ‘sloshing ions’ could have stabilized the instability by focusing pressure in regions of good-curvature near the mirrors. By also extending the end tanks to improve electron thermal confinement, 2XIIB could have achieved

classical confinement and high β , and one can speculate that a fusion gain of $Q_0 = P_{\text{fusion}}/P_{\text{Beam}} \sim 1$ would have been accessible by using the 120 keV beams employed on the TFTR tokamak in a higher field device (needed to confine the fast ions) and a larger size and density to ionize the neutral beam [4].

The WHAM mirror experiment at the University of Wisconsin-Madison will be testing these ideas in an axisymmetric, compact, high field mirror configuration. It has an intrinsically higher mirror ratio than 2XIIB due to planar coils, with a peak field of 17 T produced by the newest generation of superconducting tapes [5]. Electron heat losses are minimized by proper design of ‘expanders’ beyond the mirrors, and MHD instability is suppressed by vortex stabilization [6]. This will extend the results of the GDT (a collisional mirror experiment, characterized as gas dynamics trap) at Novosibirsk that used electron cyclotron heating to achieve the $T = 0.9$ keV [7] and $\beta = 0.6$ [8].

Unlike yin-yangs, both WHAM and a proposed future breakeven facility are axisymmetric and thus not intrinsically stable. However, axisymmetric plasma stability has been

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recently demonstrated using vortex confinement [6], while other more reactor relevant options including active feedback appear feasible [9, 10]. The high mirror ratio limit attainable with new magnets shrinks the loss cone and improves confinement, but also isotropizes the distribution function and reduces the ambipolar hole in ion velocity space that drives kinetic micro-instability.

Taking stability as given, mirror performance can be modeled with a Fokker–Planck code calculating classical losses by Coulomb collisions [11]. Section 2 gives the physics input to a new 2D (z, v) Fokker–Planck code, fast beam ion solver (FBIS), with radially averaged field $B(z)$. Stability (MHD, DCLC, AIC) can then be assessed using the ion distributions $f(v, z)$, where v is the ion speed, DCLC the drift cyclotron loss cone instability, and AIC the Alfvén ion cyclotron instability [12]. This work also generalizes to include spatial variations beyond the square well to better understand the axial electrostatic potential. The end result is a semi-analytic calculation of the ion distribution function that can be used in equilibrium and stability analysis to follow in future studies.

2. Fokker–Planck model for fast ions in a simple mirror with a cold ion population

2.1. Fokker–Planck model for $v_{th,i} \ll v \ll v_{th,e}$

We start our discussion of beam ions in magnetic mirrors by first reviewing previous results on beam ions slowing down in a square mirror dominated by a cold ion population. In this scenario the slowing down process of beam ions is typically described by the following Fokker–Planck equation

$$\frac{\partial f}{\partial t} = \frac{1}{\tau_s v^2} \frac{\partial}{\partial v} [(v^3 + v_c^3) f] + \frac{\beta_m v_c^3}{\tau_s v^3} \mathcal{L} f + S(\xi, v). \quad (1)$$

Here f is the fast ion distribution function, $\xi = v_{\parallel}/v$ is the cosine of the pitch angle,

$$\tau_s = \frac{3(2\pi)^{3/2} \epsilon_0^2 T_e^{3/2} m_f}{e^4 z_f^2 n_e \ln \Lambda_e \sqrt{m_e}} \quad (2)$$

is the Spitzer slowing down time, and m_f and z_f are the mass and charge of the fast ion, respectively [13]

$$v_c = \left(\frac{3\sqrt{\pi} m_e \ln \Lambda_i \sum_j \frac{z_j^2 n_j}{m_j}}{4n_e \ln \Lambda_e} \right)^{1/3} \left(\frac{2T_e}{m_e} \right)^{1/2} \quad (3)$$

is the critical velocity (at which the drag on the fast ions by the thermal electrons and ions is equal). Not to be confused with the normalized pressure above, we define $\beta_m = z_{\text{eff}} m_i / (2m_f)$, where z_{eff} is the effective charge of the plasma. This Fokker–Planck representation is valid for the regime $v_{ti} \ll v \ll v_{te}$, where v_{ti} and v_{te} are the ion and electron thermal speeds, respectively. In addition, the Lorentz scattering operator is defined as

$$\mathcal{L} = \frac{\partial}{\partial \xi} (1 - \xi^2) \frac{\partial}{\partial \xi}. \quad (4)$$

2.2. Eigen-functions of the scattering operator

We first seek to solve equation (1) for the ‘square’ magnetic mirror geometry

$$B = \begin{cases} B_0 & , |z| \leq L \\ R_M B_0 & , |z| > L \end{cases} \quad (5)$$

where R_M is known as the mirror ratio, and L is the half length of the central plasma cell. Neglecting the role of electric fields, in velocity space the trapped/passing boundary is characterized by $\xi_{\text{TP}} = \sqrt{1 - 1/R_M}$, and particles are trapped when $|\xi| < \xi_{\text{TP}}$.

To transform the partial differential equation in equation (1) into a set of ordinary differential equations (which are more readily integrated) we apply the well-known result that the Legendre functions P_l and Q_l are solutions to the differential equations

$$\mathcal{L} y = -\lambda y, \quad \lambda = l(l+1).$$

Because of the fast bounce motion, the solutions of interest must be even such that $M_\lambda(\xi) = M_\lambda(-\xi)$. Note that we choose to label these solutions in terms of the eigenvalue, λ , rather than the order, l . We also normalize our set of functions such that $M_\lambda(0) = 1$. For any value of λ these condition are fulfilled by a linear combination $M_\lambda = aP_l + bQ_l$, where the coefficients, a and b are obtained by solving

$$\begin{aligned} aP_l|_{\xi=0} + bQ_l|_{\xi=0} &= 1 \\ a \frac{dP_l}{d\xi} \Big|_{\xi=0} + b \frac{dQ_l}{d\xi} \Big|_{\xi=0} &= 0. \end{aligned} \quad (6)$$

As is well-known, $P_l(\xi)$ and $Q_l(\xi)$ are finite on the full interval $0 \leq \xi \leq 1$ when $l \in \{0, 1, 2, \dots\}$. However, in our case, for values of ξ within the mirror loss-cone we have $f(v, \xi) = 0$. We are therefore interested in solutions, $M_\lambda(\xi)$, only defined for the range $0 \leq \xi \leq \xi_{\text{TP}}$, and which fulfill the boundary condition that $M_\lambda(\xi_{\text{TP}}) = 0$. This boundary condition is satisfied for a discrete set of non-integers values, λ_j , which represent the eigenvalues of the problem and are readily obtained numerically. For example, table 1 provides the first eigenvalues and normalization factors for $R_M = 13.3$, similar to the WHAM experiment. The corresponding $M_{\lambda_j}(\xi)$ solutions are shown in figure 1. The functions are orthogonal on the interval defined through

$$\int_0^{\xi_{\text{TP}}} M_{\lambda_j} M_{\lambda_k} d\xi = \alpha_j \delta_{j,k}, \quad (7)$$

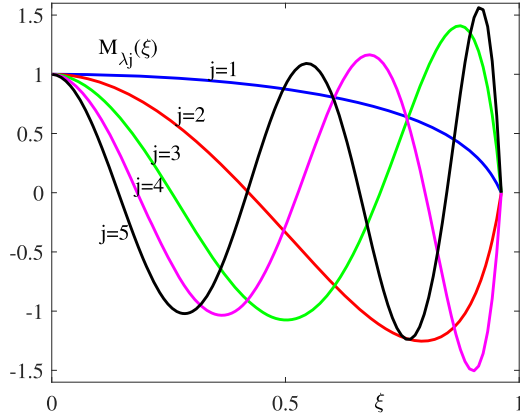
where α_j are the normalization factors,

$$\alpha_j = \int_0^{\xi_{\text{TP}}} M_{\lambda_j}^2 d\xi. \quad (8)$$

It is worth noting that the lowest eigenvalue, determined by the loss cone boundary ξ_T in the square mirror, is on the order of $2/\ln(R_M)$ as first described by [14, 15] and reviewed in [16]. More accurately, as shown in figure 2(a), based on a numerical study we find that $\lambda_1 \simeq 2/\ln(R_M) + 0.37(1/\ln(R_M))^{1.3}$.

Table 1. Eigenvalues and normalization factors of the M_{λ_j} eigen-functions for $R_M = 13.3$.

j	1	2	3	4	5	6
l_j	0.56	3.07	5.52	7.96	10.40	12.83
λ_j	0.88	12.5	36	71	118	177
α_j	0.66	0.64	0.64	0.64	0.64	0.64

**Figure 1.** Eigenfunctions of the equation $\mathcal{L}M_{\lambda_j} + \lambda_j M_{\lambda_j} = 0$ where the non-integer values of λ_j are determined from $M_{\lambda}(\xi_{TP}) = 0$.

2.3. Analytic solution for f

The eigenfunctions form a complete set for $0 \leq \xi \leq \xi_{TP}$, allowing any distribution function $f(v, \xi)$ to be expanded in the form given by

$$f(v, \xi) = \sum_j f_j(v) M_{\lambda_j}(\xi). \quad (9)$$

The anisotropic source is assumed to be monoenergetic $4\pi v^2 \langle S \rangle = S_0 \delta(v - v_0)$, where v_0 is the injection velocity and we use the notation $S = dN/(dv d\mathbf{x} dt)$ and $S_0(\xi) = dN/(d\mathbf{x} dt)$. For a given beam injection geometry, $S_0(\xi)$ can be determined as outlined in reference [17]. Assuming the source is known, it can be expanded in the basis functions

$$\frac{S_0}{4\pi} = \sum_j S_j M_{\lambda_j}(\xi), \quad (10)$$

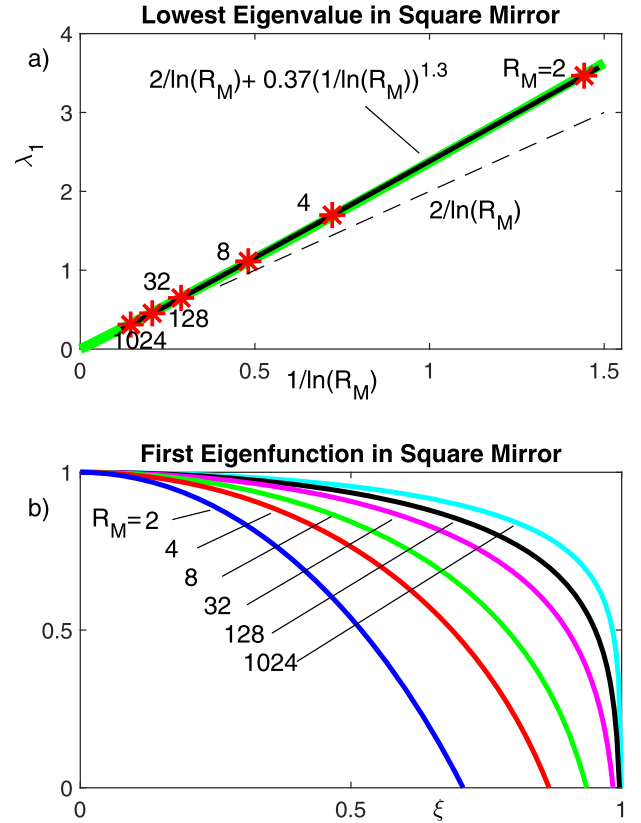
where the expansion factors are determined as

$$S_j = \frac{1}{4\pi \alpha_j} \int_0^{\xi_{TP}} S_0 M_{\lambda_j} d\xi. \quad (11)$$

Below we find that in a mirror-fusion device, the $j = 1$ term will dominate the ion distribution. For the case where $S_0(\xi)$ is peaked where $M_{\lambda_j}(\xi) \simeq 1$, we then have

$$S_1 = \frac{1}{4\pi \alpha_1} \frac{I_{\text{beam}}/e}{V}, \quad (12)$$

where I_{beam} is the current of the neutral beam, and V is the volume of the device.

**Figure 2.** (a) The black line represents the lowest eigenvalue of $\mathcal{L}M = -\lambda M$ obtained with the boundary condition $M(\xi_T) = M(\sqrt{1 - 1/R_M}) = 0$, plotted as a function of $1/\ln(R_M)$. The green line is the approximation indicated. Values of λ_1 are marked, corresponding to particular selections of R_M . (b) Eigenfunctions for the values of R_M selected in (a).

Using $\mathcal{L}M_{\lambda_j} = -\lambda_j M_{\lambda_j}$, equation (1) now reduces to a set of ordinary differential equations

$$-\frac{\beta_m v_c^3}{v^3} \lambda_j f_j + \frac{1}{v^2} \frac{\partial}{\partial v} [(v^3 + v_c^3) f_j] = -\tau_s S_j \frac{\delta(v - v_0)}{v^2}, \quad (13)$$

and with these definitions the full solution for $v < v_0$ becomes

$$f(v, \xi) = \frac{\tau_s}{v^3 + v_c^3} \sum_j S_j M_{\lambda_j}(\xi) u^{\lambda_j}, \quad (14)$$

where

$$u = \left(\frac{v_0^3 + v_c^3}{v^3 + v_c^3} \right)^{\beta_m/3}. \quad (15)$$

From equation (14) it is clear that the critical velocity, v_c is central to the beam ion dynamics. Considering a pure 50/50 DT fusion plasma v_c defined in equation (3) may also be written in terms of the critical energy

$$\mathcal{E}_c \equiv \frac{1}{4} (m_D + m_T) v_c^2 \simeq 20 T_e. \quad (16)$$

An example of the solution in equation (14) is shown in figure 3.

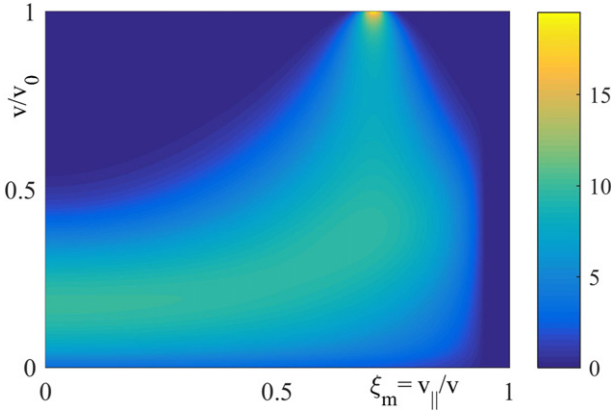


Figure 3. Beam ion distribution function calculated using equation (14), for a 60 keV beam at 45° relative to $\mathbf{B}$ into a 1 keV plasma, for a GDT with a mirror ratio $R_M = 13.3$.

3. Reduced model for a pure beam plasma

We now consider another important case where the beam ions themselves are the background plasma for the slowing down process. For simplicity, we will for now model the D and T beam distributions as one common beam distribution with mass $m_f = (m_D + m_T)/2$. Using the square mirror assumption, in this preliminary analysis we also neglect the effects of spatially distributed electrostatic potential, $\Phi_{\parallel} \simeq 5T_e$ [18], but the role of $\Phi_{\parallel}$ is included in the power-balance of the electrons. These approximations make the problem analytically tractable and will provide a basic model yielding a quantitative understanding of the physical processes influencing the performance of the mirror. In later sections we compare numerical results from a more complete description with the result from the basic model.

3.1. Fundamental conditions on the fusion performance of the magnetic mirror

First, without specifying a particular model for the beam ion behavior, we may obtain some general results related to the normalized fusion yield of the magnetic mirror, $Q_0 = P_{\text{fusion}}/P_{\text{beam}}$ (throughout the paper we will assume that each DT fusion reaction creates 22.4 MeV of energy, here including the energy gained by breeding of tritium in a lithium blanket). For the present square mirror with one common beam distribution we model the fusion power as

$$P_{\text{fusion}} = n_b^2 V \Delta \mathcal{E}_{\text{fusion}} \frac{1}{4} \langle v\sigma \rangle_{\text{DT}}.$$

Here $\langle v\sigma \rangle_{\text{DT}}$ is the DT fusion cross-section, often tabulated as a function of a common ion temperature. Meanwhile, the beampower is naturally $P_{\text{beam}} = \mathcal{E}_{\text{beam}} I_{\text{beam}}/e$. It is convenient to introduce the particle confinement time, τ_P , defined such that $Vn_b \equiv \tau_P I_{\text{beam}}/e$. Furthermore, we will use the 90°-pitch angle scattering time evaluated at the effective beam temperature $\tau_{T_i}^{90} = 1/(n_b v_{ti} \sigma_{v_{ti}}^{ii})$, where $v_{ti} \sigma_{v_{ti}}^{ii}$ then represents the ‘scattering reactivity’. With these definitions we readily obtain

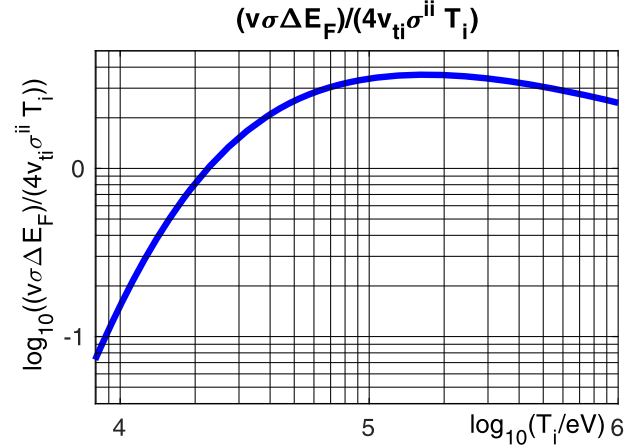


Figure 4. Normalized fusion reactivity $\langle v\sigma \rangle/(4v_{ti} \sigma_{v_{ti}}^{ii} T_i)$ for Maxwellian distributed beam ions.

$$\frac{Q_0}{\tilde{T}_i \tilde{\tau}_P} = \frac{\Delta \mathcal{E}_{\text{fusion}}}{T_i} \frac{\langle v\sigma \rangle_{\text{DT}}}{4v_{ti} \sigma_{v_{ti}}^{ii}}, \quad (17)$$

where

$$\tilde{\tau}_P = \frac{\tau_P}{\tau_{T_i}^{90}}, \quad \tilde{T}_i = \frac{T_i}{\mathcal{E}_{\text{beam}}} \quad \text{and} \quad \tilde{T}_e = \frac{T_e}{\mathcal{E}_{\text{beam}}}. \quad (18)$$

For a Maxwellian distribution the rhs of equation (17) is a function of only the temperature T_i . It is a dimensionless quantity which measures the fusion energy gain for each 90°-scattering-event and is then normalized by T_i . As shown in figure 4 it has a maximum of ~ 3.6 for $T_i \simeq 140$ keV, which is the effective ion temperature that optimizes the fusion yield, Q_0 . We will find that even for the non-Maxwellian distributions expected in a mirror reactor this form is still accurate when evaluated as a function of the effective temperature of the beam ions. Thus, this function is general to mirror devices and represents a fundamental boundary condition for magnetic mirror fusion imposed by nature.

The denominator on the lhs of equation (17) has two terms, $\tilde{T}_i$ and $\tilde{\tau}_P$, such that the value of Q_0 is linear in both. This is consistent with the intuitive result that the performance of the device increases with the normalized ion temperature and the normalized particle confinement time. Note that the dimensionless $\tilde{\tau}_P$ represents the number of 90°-scattering-events it takes the average ion before its random walk is terminated in the losscone. Below we will find that $\tilde{\tau}_P \simeq 1.5 \log_{10}(R_M)$, increasing slowly with an increasing mirror ratio. Again, these results are general, independent of the particular model imposed for characterizing the beam ions. In addition, for the Fokker–Planck models considered here, both $\tilde{T}_i$ and $\tilde{\tau}_P$ turn out to be independent of the absolute value of T_i . This makes equation (17) particularly useful for determining T_i , as then only the rhs depends on this effective temperature.

3.2. Approximate form of $f_1(v)$

We now seek to update the Fokker–Planck model to a form that is consistent with a pure beam plasma. In equation (13) the term $(1/v^2) \partial v_c^3 f_j / \partial v$ represents the drag on the beam ions

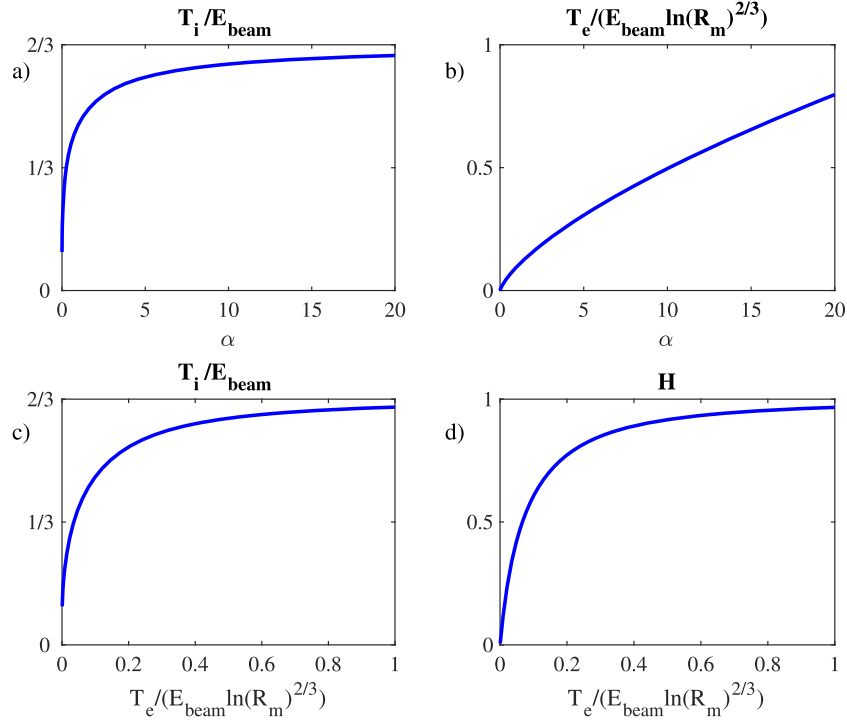


Figure 5. (a) $\tilde{T}_i \equiv T_i / \mathcal{E}_{\text{beam}}$ as a function of α in equation (21). (b) Similarly, $\tilde{T}_e / \ln(R_M)^{2/3}$ as a function of α . (c) $\tilde{T}_i$ as a function of $\tilde{T}_e / \ln(R_M)^{2/3}$. (d) The function $\mathcal{H}$ in equation (23) as a function of $\tilde{T}_e / \ln(R_M)^{2/3}$.

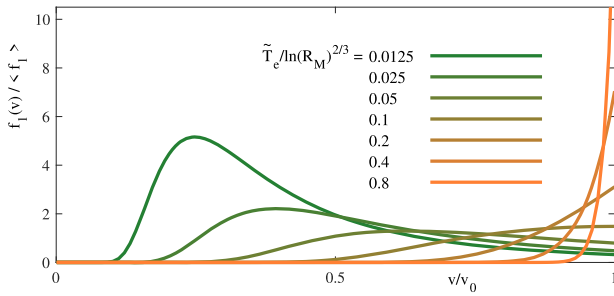


Figure 6. Normalized distributions $f_1(v)/\langle f_1 \rangle = v_0^3 f_1(v) / (3 \int v^2 f_1 dv)$, for various values of $\tilde{T}_e / \ln(R_M)^{2/3}$.

from a cold ion population. For the present case of a pure beam plasma without a cold ion population, this term should be neglected. Furthermore, in a reactor scenario the critical velocity normalized to the beam injection velocity will be relatively large, $v_c/v_0 \simeq 1$, such that the beam ion dynamics is dominated by scattering off ions. In this case, because $\lambda_1 \ll \lambda_j$ with $j \in \{2, 3, \dots\}$, by inspection of equation (13) it follows that $f_1 \gg f_j$, such that the beam ion distribution will be dominated by the eigenfunction $M_{\lambda_1}(\xi)$. The equation for f_1 is then approximately given by

$$-\frac{\beta_m v_c^3}{v^3} \frac{\partial g}{\partial v} \lambda_1 f_1 + \frac{1}{v^2} \frac{\partial}{\partial v} v^3 f_1 = -\tau_s S_1 \frac{\delta(v - v_0)}{v^2}, \quad (19)$$

where we have introduced the density normalized Rosenbluth potential

$$g(v) = \int f_1(v') |\mathbf{v} - \mathbf{v}'| d^3 v' / \int f_1(v') d^3 v'.$$

In principle, g should be computed using the solution f_1 of equation (19). However, this solution is not available *a priori*, and for the purpose of evaluating g we use $\partial g / \partial v \simeq v / (2v_{ti})$ representative for a Maxwellian f_1 with temperature $T_i = mv_{ti}^2/2$. Integration of equation (19) then yields

$$f_1 = \begin{cases} \frac{\tau_s S_1}{\exp(-\alpha)} \frac{\exp(-\alpha v_0^2/v^2)}{v^3} & \text{for } v \leq v_0, \\ 0 & \text{for } v > v_0 \end{cases} \quad (20)$$

where

$$\alpha \equiv \frac{v_c^3 \beta_m \lambda_1}{4v_0^2 v_{ti}} \simeq 22.4 \frac{\tilde{T}_e^{3/2}}{\tilde{T}_i^{1/2} \ln(R_M)}. \quad (21)$$

In this last approximation, we have used the fact that for the square mirror the lowest eigenvalue is $\lambda_1 \simeq 2/\ln(R_M)$, evident from figure 2(a).

The analytical form for f_1 is useful as it permits an analysis of the dependencies of the effective ion temperature, plasma density and fusion reactivity. The full distribution is $f \simeq f_1(v) M_1(\xi)$, and ignoring the ξ dependency we first estimate the effective ion temperature

$$\frac{3}{2} \frac{T_i}{\mathcal{E}_{\text{beam}}} = \frac{\int_0^{v_0} v^4 f_1 dv}{\int_0^{v_0} v^2 v_0^2 f_1 dv} = \frac{\exp(-\alpha) - \Gamma(0, \alpha)}{\Gamma(0, \alpha)}, \quad (22)$$

where $\Gamma(0, \alpha)$ is the incomplete gamma function. Because the beam distributions generally are non-Maxwellian, $\tilde{T}_i$ is not an

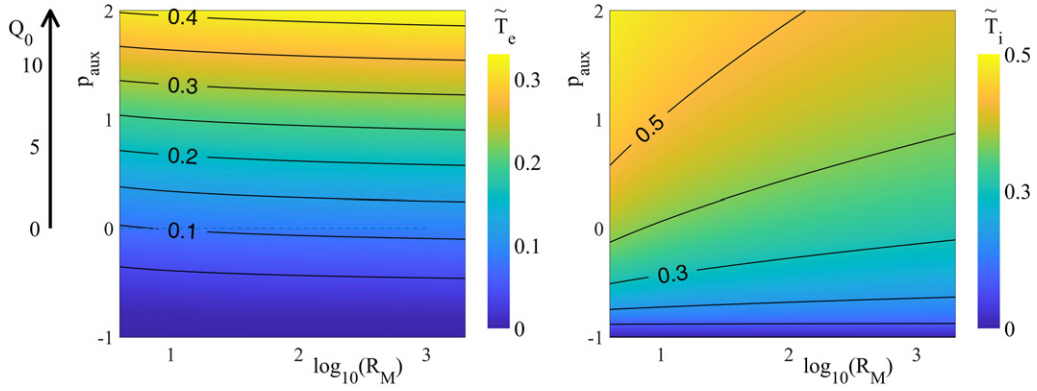


Figure 7. Predictions of $\tilde{T}_e$ and $\tilde{T}_i$ as functions of $\log_{10}(R_M)$ and auxiliary electron heating $p_{aux} = (P_\alpha + P_{other})/\mathcal{E}_{beam}$. As indicated by the Q_0 -axis on the left, when the sole auxiliary power is from alpha particle heating, we have $p_{aux} \simeq Q_0/6.7$.

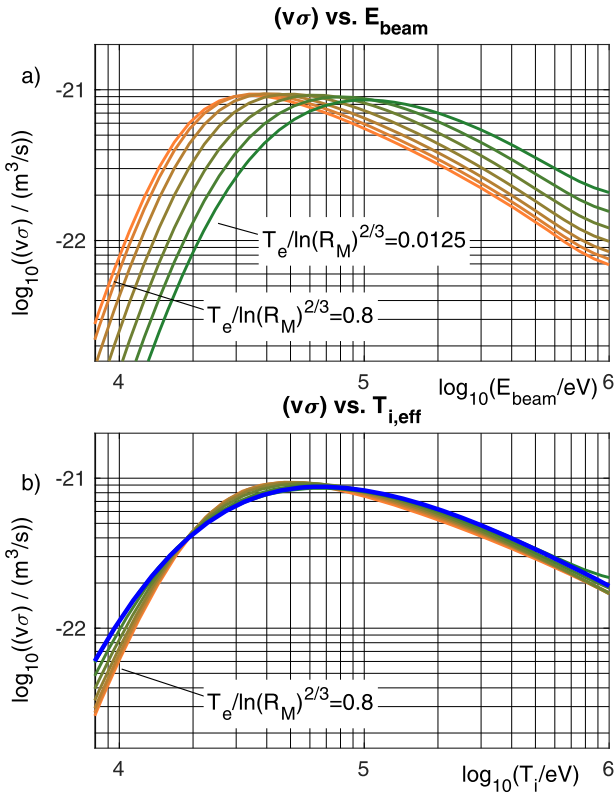


Figure 8. (a) Fusion reactivity, $\langle v\sigma \rangle$, as a function of $\mathcal{E}_{beam}$, for $\tilde{T}_e/\ln(R_M)^{2/3} \in \{0.0125, 0.025, 0.05, 0.1, 0.2, 0.4, 0.8\}$. (b) Fusion reactivity, $\langle v\sigma \rangle$, as a function of T_i . The blue line is the result for Maxwellian beam ions, representing a good approximation (error less than 10%) for the expected values of $\tilde{T}_e/\ln(R_M)^{2/3} \simeq 0.02$.

actual temperature, but should merely be interpreted as the characteristic energy of the beam distribution. In figure 5(a) the profile of $\tilde{T}_i = T_i/\mathcal{E}_{beam}$ is shown as a function of α . Given $\tilde{T}_i(\alpha)$ is now known, we readily obtain $\tilde{T}_e/\ln(R_M)^{2/3}$ also as a function of α . This in turn permits us to recast $\tilde{T}_i$ as a function of $\tilde{T}_e/\ln(R_M)^{2/3}$. The profiles of $\tilde{T}_e/\ln(R_M)^{2/3}$ as a function of α , and $\tilde{T}_i$ as a function of $\tilde{T}_e/\ln(R_M)^{2/3}$ are shown in figures 5(b) and (c), respectively. To be useful later for calculating the beam ion density, figure 5(d) shows a function $\mathcal{H}$ defined as

$$\mathcal{H}(\alpha) \equiv \alpha \frac{\Gamma(0, \alpha)}{e^{-\alpha}} \simeq 1 - \frac{1}{\alpha} + \dots, \quad (23)$$

approaching unity for large α .

With equation (20), and the knowledge of α as a function of $\tilde{T}_e/\ln(R_M)^{2/3}$ (see figure 5(b)), the normalized distributions $f_1/\langle f_1 \rangle = f_1(v)/(4\pi v_0^3/3)/(4\pi \int v^2 f_1 dv)$ are readily obtained and are shown in figure 6 for a range of $\tilde{T}_e/\ln(R_M)^{2/3}$. At low $\tilde{T}_e/\ln(R_M)^{2/3}$ values, electron drag is strong causing the ions to slow down significantly. In the opposite limit we find $f_1 \propto \delta(v - v_0)$, such that $\tilde{T}_i \simeq 2/3$. In this latter scenario the change in the speed of the beam ions is negligible during their process of pitch angle diffusion into the loss cone. For fixed $\tilde{T}_e$, an increase in the mirror ratio R_M will cause a reduction in $\tilde{T}_e/\ln(R_M)^{2/3}$ and therefore also a reduction in $\tilde{T}_i$. This effect may appear counter intuitive, but occurs because an enhanced value of R_M increases the confinement time and thereby increases the work done by electron drag.

3.3. Self-consistent values of $\tilde{T}_e$ and $\tilde{T}_i$

Above we obtained equations permitting an evaluation of $\tilde{T}_i$ as a function of $(R_M, \tilde{T}_e)$. We next seek to determine the value of $\tilde{T}_e$ that is consistent with power-balance of the device. For beam heating, the input power per particle is naturally $\mathcal{E}_{beam}$. In addition, we also include the possibility that the average electron is heated by an auxiliary source $p_{aux} = (P_\alpha + P_{other} + P_{loss})/\mathcal{E}_{beam}$ including P_α of electron heating from the slow down of fusion α -particles. The contributions through P_{other} and P_{loss} can account for, say, auxiliary RF-heating, plasma heat-losses ($P_{loss} < 0$) possibly induced by plasma turbulence, ionization of neutrals by electrons, etc. For $P_{other} = P_{loss} = 0$, we have $\tilde{p}_{aux} = Q_0 \mathcal{E}_\alpha/\mathcal{E}_{fusion}$, where again $Q_0 = P_{fusion}/P_{beam}$ is the fusion yield of the device.

In the present beam-heating problem the inevitable energy losses are caused by electrons and ions exiting the device through the mirror throat. We assume each escaping electron carries an energy of T_e , while each ion approximately carries $T_i + e\Phi_w$. Here, $e\Phi_w \simeq 5T_e$ is the electrostatic potential that self-consistently develops in the mirror to ensure that the electrons and the ions leave the device at the same rate [18];

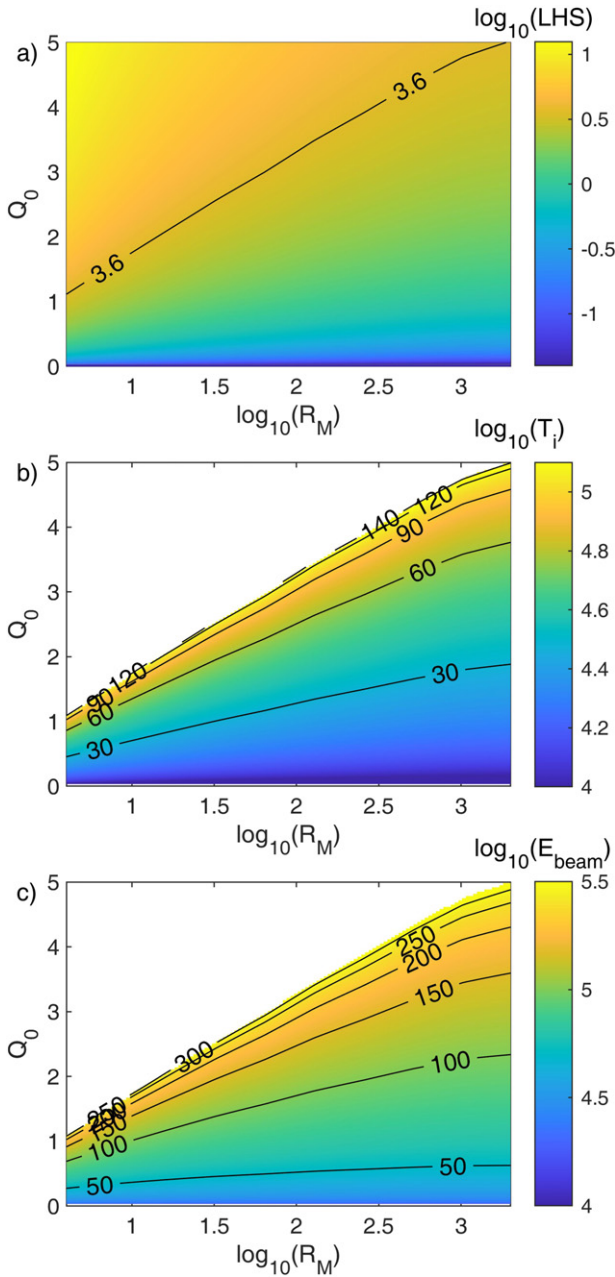


Figure 9. (a) Color contours of $\log_{10}(\text{lhs})$, where lhs is the left-hand side of equation (32). (b) Values of $\log_{10}(\text{lhs}) < \log_{10}(3.6)$ are within the available range of $\langle v\sigma \rangle / (4v_{ti}\sigma_{v_{ti}}^{\text{ii}} T_i)$ in figure 4, allowing T_i to be inferred. In turn, $\mathcal{E}_{\text{beam}} = T_i/\tilde{T}_i$, shown in (c), is then readily computed.

only very fast electrons are able to escape over this confining potential.

Equating the input power with the identified losses and normalizing by $\mathcal{E}_{\text{beam}}$ we obtain

$$1 + \tilde{p}_{\text{aux}} = \tilde{T}_i(\tilde{T}_e, R_M) + 6\tilde{T}_e. \quad (24)$$

Given the knowledge of $\tilde{T}_i(\tilde{T}_e, R_M)$, as displayed in figure 5, we can readily solve equation (24) and obtain the self-consistent values of $\tilde{T}_e$ and $\tilde{T}_i$ as functions of $(\tilde{p}_{\text{aux}}, R_M)$. These profiles displayed in figure 7 are remarkable as they, for a given R_M , allow us to predict the temperature of the plasma

for any set of beam energies and the extra auxiliary heating fraction. For example, for $\tilde{p}_{\text{aux}} = 0$ we find that $\tilde{T}_e \simeq 0.1$ nearly independent of R_M , whereas $\tilde{T}_i \simeq 0.4$ for $\log_{10}(R_M) \simeq 1$ and decreases to $\tilde{T}_i \simeq 0.35$ as $\log_{10}(R_M) \simeq 3$ is approached. These results are independent of I_{beam} as well as the plasma density in the device.

3.4. The particle confinement time, τ_P

Based on the above framework, the particle confinement time can also be estimated. Using $f \simeq f_1(v)M_1(\xi)$ with equation (20) we evaluate the beam-ion density

$$\begin{aligned} n_b &= 4\pi \int_0^{v_0} v^2 f_1 dv \int_0^1 M_1(\xi) d\xi \\ &= \frac{4\pi\tau_s S_1}{e^{-\alpha}} \int_0^1 \frac{e^{-\alpha/y^2}}{y} dy \int_0^1 M_1(\xi) d\xi \\ &= \frac{2\pi\tau_s S_1}{\alpha} \mathcal{H}(\alpha) \int_0^1 M_1(\xi) d\xi, \end{aligned} \quad (25)$$

where $\mathcal{H}$ was introduced in equation (23) and shown in figure 5(d).

The right-hand side of equation (25) is a function of the plasma density through the factor $\tau_s/\alpha = (4v_{ti}/(\lambda_1 v_0))(\tau_s v_0^3/\beta_m v_c^3)$, and equation (25) therefore does not represent a closed form for evaluating n_b . Meanwhile, we notice that $\tau_s v_0^3/(\beta_m v_c^3) = \tau_0^{\text{ii}}$ is the ion scattering time at the injection speed $v = v_0$. This may also be written in terms of the scattering cross-section (at v_0) such that $\tau_s v_0^3/(\beta_m v_c^3) = 1/(n_b v_0 \sigma_{v_0}^{\text{ii}})$, making the inverse density dependence explicit. Applying this substitution in equation (25) and solving for n_b we find

$$n_b = \sqrt{\frac{4\pi S_1 v_{ti}}{\lambda_1 v_0} \mathcal{H} \frac{1}{v_0 \sigma_{v_0}^{\text{ii}}} \int_0^1 M_1(\xi) d\xi}. \quad (26)$$

The particle confinement time $\tau_P = n_b V/(I_{\text{beam}}/e)$ is governed by collision and we expect that $n_b \tau_P$ is independent of n_b . Indeed, because $I_{\text{beam}} = 4\pi\alpha_1 e S_1 V$ (see equation (12)) and using equation (26) we obtained

$$n_b \tau_P = \frac{n_b^2 V}{I_{\text{beam}}/e} = \frac{v_{ti}}{\alpha_1 \lambda_1 v_0} \frac{\mathcal{H}}{v_0 \sigma_{v_0}^{\text{ii}}} \int_0^1 M_1(\xi) d\xi. \quad (27)$$

Above we found an expression for $\tilde{T}_i$, and to evaluate Q_0 in equation (17), we now only need to determine the normalized confinement time introduced in equation (18). For this, we notice that

$$\frac{v_0 \sigma_0^{\text{ii}}}{\tilde{T}_i^{3/2}} = \frac{e^4 \beta_m \log \Lambda_i}{4\pi \epsilon_0^2 m_f^2 v_{ti}^3} \equiv v_{ti} \sigma_{v_{ti}}^{\text{ii}}, \quad (28)$$

where again $v_{ti} \sigma_{v_{ti}}^{\text{ii}}$ is the ‘scattering-reactivity’ evaluated at v_{ti} . Using equation (27) in combination with $\tau_{Ti}^0 = 1/(n_b v_{ti} \sigma_{v_{ti}}^{\text{ii}})$ it then follows that

$$\tilde{\tau}_P = \frac{1}{\alpha_1 \lambda_1} \frac{\mathcal{H}}{\tilde{T}_i} \int_0^1 M_1(\xi) d\xi. \quad (29)$$

Furthermore, as a useful approximation, using the results in figures 5(c) and (d) it is found that for the relevant values of $T_e/\ln(R_M)^{2/3} \simeq 0.02$ we have $\tilde{T}_i \simeq \mathcal{H}$, such that

$$\tilde{\tau}_P \simeq \frac{1}{\lambda_1} \frac{\int M_1 d\xi}{\int M_1^2 d\xi} \simeq \frac{1.3}{\lambda_1} \simeq \frac{3}{2} \log_{10}(R_M). \quad (30)$$

Given $\tau_{ii} \text{ (s)}^{-1} \simeq 6.5 \times 10^{-4} T_{i,\text{keV}}^{3/2}/n_{20}$, and $\tilde{T}_i \simeq 0.4$, this form is consistent with previous results [12]

$$\tau_P = \mathcal{C} \mathcal{E}_{\text{beam,keV}}^{3/2} \log_{10}(R_M)/n_{20}, \quad (31)$$

where $\mathcal{C}(R_M, p_{\text{aux}}) \simeq 2.5\text{--}2.8 \times 10^{-4} \text{ (s)}$.

3.5. Fusion performance of the ideal square magnetic mirror

With $\tilde{T}_i$ known from figure 7(b) and $\tilde{\tau}_P$ given in equation (29), we are now nearly ready to evaluate Q_0 in equation (17). However, it remains to be shown that the rhs of equation (17) is well represented by the results in figure 4 obtained for Maxwellian ions. For this, in figures 8(a) and (b), for the same values of $\tilde{T}_e/\ln(R_M)^{2/3}$ as used in figure 6, the profiles of $\langle v\sigma \rangle$ are shown as functions of $\mathcal{E}_{\text{beam}}$ and T_i , respectively. For the expected values of $\tilde{T}_e/\ln(R_M)^{2/3} \simeq 0.02$ the fusion reactivities are well approximated by the results for Maxwellian distributed ions (blue line), and the result in figure 4 is therefore representative of the rhs of equation (17).

To emphasize the dependency of Q_0 on the first eigenvalue $\lambda_1 \simeq 2/\ln(R_M)$ we recast equation (17) as

$$Q_0 \lambda_1 F = \frac{\Delta \mathcal{E}_{\text{fusion}}}{T_i} \frac{\langle v\sigma \rangle}{4v_{ti} \sigma_{v_{ti}}^{ii}}, \quad (32)$$

with

$$F(Q_0, R_M) \equiv \frac{\int M_1^2 d\xi}{\mathcal{H} \int M_1 d\xi}. \quad (33)$$

The lhs of equation (32) is a function of only (Q_0, R_M) and is shown in figure 9(a). By solving equation (32) numerically, the absolute effective ion temperature T_i is readily determined as shown in figure 9(b). In turn, for the case without additional heating or energy losses $\tilde{p}_{\text{aux}} = Q_0 \mathcal{E}_\alpha / \mathcal{E}_{\text{fusion}}$ allowing $\mathcal{E}_{\text{beam}}(Q_0, R_M) = T_i / \tilde{T}_i$ to be determined and is shown in figure 9(c).

4. Orbit averaged Lorentz operator for non-uniform B

In the analysis above we considered the square mirror. Of course, a perfectly square magnetic well is not realistic and it is useful to understand the impact on the beam ions for experimental scenarios which can actually be implemented in the laboratory. In the following we therefore develop a mathematical framework which permits the above analysis to be generalized to arbitrary magnetic configurations.

4.1. Constant of motion variables

For a non-uniform magnetic field the pitch angle varies along the ion trajectories, and $\xi = v_{\parallel}/v$ is no longer an optimal

variable for describing the beam ion dynamics. Meanwhile, in the fast transit time limit, the beam ion distribution function becomes independent of the ignorable coordinates describing the actual location of the ions along their near instantaneous drift orbits. The distribution function can then be expressed in terms of constant of motion variables:

$$f = f(U, \mu, \psi, t). \quad (34)$$

Here $U = \mathcal{E} + e\Phi$ is the total particle energy and $\mathcal{E} = mv^2/2$ is the kinetic energy, $\mu = mv_{\perp}^2/(2B)$ is the magnetic moment, while

$$\psi(r, z) = \int_0^r r' B_z(z, r') dr'$$

is the magnetic flux function. In the section above we were able to find solutions separable in the variables $\mathcal{E}$ and ξ . For this it was fundamental to impose $\Phi = 0$ within the square well. Similarly, here we will also impose $\Phi = 0$, such that $U = \mathcal{E}$. In the subsequent numerical analysis we will show that $e\Phi/\mathcal{E}_{\text{beam}} \simeq 0.1\text{--}0.2$, and the effect of Φ can be accounted for as a small correction, mostly affecting the lowest energy ions of the idealized numerical solution.

In place of μ , we find it is useful to introduce the pitch angle variable

$$\Lambda = \frac{\mu B_0}{\mathcal{E}} = \frac{\mathcal{E}_{\perp}}{\mathcal{E}} \frac{B_0}{B}. \quad (35)$$

For a configuration with minimum magnetic field strength $B_{\text{min}} = B_0$ and maximum B_{max} , the lowest value of Λ is 0 and occurs when $\mathcal{E}_{\perp} = 0$. The largest possible value of Λ is $\Lambda_{\text{max}} = B_0/B_{\text{min}} = 1$, which is found for orbits which have $\mathcal{E}_{\perp} = \mathcal{E}$ at the location where B is at its minimum. These are the orbits which are localized at the mid-plane with $v_{\parallel} = 0$. We then have $0 \leq \Lambda \leq 1$. Particles are magnetically trapped when $\mu B_{\text{max}} \geq \mathcal{E}$, which may also be written as $\Lambda \geq 1/R_M$. Thus, the trapped range in Λ is $1/R_M \leq \Lambda \leq 1$.

Below we will also make use of the orbit bounce time,

$$\tau_b = \oint \frac{1}{v_{\parallel}} dl = 4 \int_0^{L_b} \frac{1}{v_{\parallel}} dl, \quad (36)$$

and the normalized bounce time

$$\tilde{\tau}_b = \frac{v}{4l} \oint \frac{1}{v_{\parallel}} dl = \frac{1}{l} \int_0^{L_b} \frac{v}{v_{\parallel}} dl. \quad (37)$$

In these $z = L_b$ marks the turning point in the orbit bounce motion.

4.2. Orbit averaging of the Lorentz operator

We now seek to recast the Fokker–Planck description in terms of the constant of motion variables. Here, a main concern is that $\mathcal{L}$ in equation (4) is expressed in terms of ξ and not in terms of a constant of motion. Locally we can assume that the scattering process is still described by $\mathcal{L}$, which is then subject to orbit averaging [13, 19]. For this orbit averaging, it is convenient to express $\mathcal{L}$ in terms of Λ . Starting with $\mathcal{L}$ in equation (4)

$$\mathcal{L} = \frac{\partial}{\partial \xi} (1 - \xi^2) \frac{\partial}{\partial \xi} \quad (38)$$

we rewrite the scattering operator in terms of the new variables, Λ and $\mathcal{E}$

$$\mathcal{L} = \frac{mv_{\parallel}B_0^2}{\mathcal{E}^2B} \frac{\partial}{\partial \Lambda} \bigg|_{\mathcal{E}} \mu v_{\parallel} \frac{\partial}{\partial \Lambda} \bigg|_{\mathcal{E}}, \quad v_{\parallel} = \sqrt{2\mathcal{E}(1 - \Lambda \tilde{B})/m} \quad (39)$$

which then takes the form

$$\mathcal{L} = \left(\frac{4}{\tilde{B}} - 6\Lambda \right) \frac{\partial}{\partial \Lambda} \bigg|_{\mathcal{E}} + \frac{4}{\tilde{B}} \Lambda (1 - \Lambda \tilde{B}) \frac{\partial^2}{\partial \Lambda^2} \bigg|_{\mathcal{E}}. \quad (40)$$

We then average over a bounce motion. Using $dt = dz/v_{\parallel}$, this time average can be expressed as a spatial average along the length of the mirror

$$\langle \mathcal{L} \rangle_x = \left(4 \left\langle \frac{1}{\tilde{B}} \right\rangle_z - 6\Lambda \right) \frac{\partial}{\partial \Lambda} \bigg|_{\mathcal{E}} + 4\Lambda \left(\left\langle \frac{1}{\tilde{B}} \right\rangle_z - \Lambda \right) \frac{\partial^2}{\partial \Lambda^2} \bigg|_{\mathcal{E}}, \quad (41)$$

where the spatial averaging $\langle (\dots) \rangle_z$ is defined as

$$\langle (\dots) \rangle_z = \frac{1}{l \tilde{\tau}_b} \int_0^{L_b} \frac{(\dots)}{\sqrt{1 - \Lambda \tilde{B}}} \frac{n(z)}{\langle n \rangle} dz. \quad (42)$$

Here, under the integral we have included the factor $n(z)/\langle n \rangle$, because the scattering frequency scales linearly with the plasma density (see equations (1) and (2)). Below we will show how the spatial variations of $n(z)$ become important. With equation (41) the orbit average of $\mathcal{L}$ can now be calculated as a function of Λ for any magnetic profile $B(z)$.

4.3. New pitch angle variable

In the square mirror the total number of particles can be calculated as

$$N = V \int 4\pi v^2 dv \int_0^1 f(v, \xi) d\xi. \quad (43)$$

The last integral over ξ includes no Jacobian factor and this fact is a fundamental property also tied to the form of the expansion in equation (9), as well as the orthogonality condition in equation (7). In the following, for a general magnetic geometry $B(z)$, we therefore seek a constant of motion variable η in terms of which the density can be calculated by a form similar to equation (43). After identifying η , we can then introduce an expansion similar to equation (9) also for the general magnetic mirror with non-uniform magnetic fields.

For our present set of constant of motion variables, the relative size of the phase space volumes is given by

$$\mathcal{J} \equiv \frac{\Delta \mathbf{x} \Delta \mathbf{v}}{\Delta \mathcal{E} \Delta \mu \Delta \psi} = \frac{4\pi^2 \tau_b}{m^2}. \quad (44)$$

This result is obtained in references [17, 20]. The total number of particles within a plasma column of length $2L$ can then be calculated as

$$\begin{aligned} N &= \int f d^3v d^3x = \frac{4\pi^2}{m^2} \int d\mathcal{E} \int d\mu \int d\psi \tau_b f, \\ &= \left(\frac{4\pi L}{B_0} \int d\psi \right) \int 4\pi v^2 dv \int d\Lambda \tilde{\tau}_b f(v, \Lambda). \end{aligned}$$

We note that the volume of the plasma is

$$V = \langle 1 \rangle_{\Lambda} \frac{4\pi L}{B_0} \int d\psi,$$

where we apply the notation of reference [21]:

$$\langle 1 \rangle_{\Lambda} \equiv \int_0^1 d\Lambda \tilde{\tau}_b. \quad (45)$$

The total number of particles is then given by

$$N = V \int 4\pi v^2 dv \frac{1}{\langle 1 \rangle_{\Lambda}} \int d\Lambda \tilde{\tau}_b f(v, \Lambda). \quad (46)$$

We now introduce a new constant of motion variable

$$\eta = \eta(\Lambda) = 1 - \frac{1}{\langle 1 \rangle_{\Lambda}} \int_0^{\Lambda} d\Lambda' \tilde{\tau}_b, \quad (47)$$

where it is clear that $0 \leq \eta \leq 1$. We also notice that

$$\frac{\partial \eta}{\partial \Lambda} = -\frac{\tilde{\tau}_b}{\langle 1 \rangle_{\Lambda}},$$

and similar to equation (43), the expression for the total number of particles then becomes

$$N = V \int 4\pi v^2 dv \int_0^1 f(v, \eta) d\eta. \quad (48)$$

For any magnetic configuration, $\tilde{\tau}_b(\Lambda)$ in equation (37) can be computed, and equation (47) is readily integrated to provide η as a function of Λ . As will be discussed further below, examples of $\eta(\Lambda)$ are shown in figure 10(b).

To further demonstrate the properties of η , it is useful to consider the square mirror for which $\tilde{\tau}_b = v/v_{\parallel}$ and $\Lambda = v_{\perp}^2/v^2$. Using $\tilde{v}_{\perp} = v_{\perp}/v$, and $\tilde{\tau}_b = 1/\sqrt{1 - \tilde{v}_{\perp}^2}$ we find

$$\eta = 1 - \frac{\int_0^{\tilde{v}_{\perp}^2} 1/\sqrt{1 - \tilde{v}_{\perp}^{\prime 2}} d\tilde{v}_{\perp}^{\prime}}{\int_0^1 1/\sqrt{1 - \tilde{v}_{\perp}^{\prime 2}} d\tilde{v}_{\perp}^{\prime}} = \frac{|v_{\parallel}|}{v}. \quad (49)$$

Thus, for the uniform flux tube we have $\eta = |v_{\parallel}|/v$, while for the general case η can be considered a representative value of $|v_{\parallel}|/v$ along the orbit. The constant of motion variable η is thus a generalization of ξ used in the square mirror above, and η will be useful for generalizing the results for the square mirror to more general magnetic mirror geometries.

4.4. Eigenfunctions of the orbit averaged Lorentz operator $\langle \mathcal{L} \rangle_z$

In this section we will be concerned with obtaining the eigenfunctions for the orbit average Lorentz operator. Thus, our aim is to obtain the set of functions $I_j(\eta)$ that fulfill the following eigenvalue equation

$$\langle \mathcal{L} \rangle_z I_j(\eta) = -\lambda_j I_j(\eta), \quad (50)$$

where λ_j is the equivalent to λ_j found for the square mirror above.

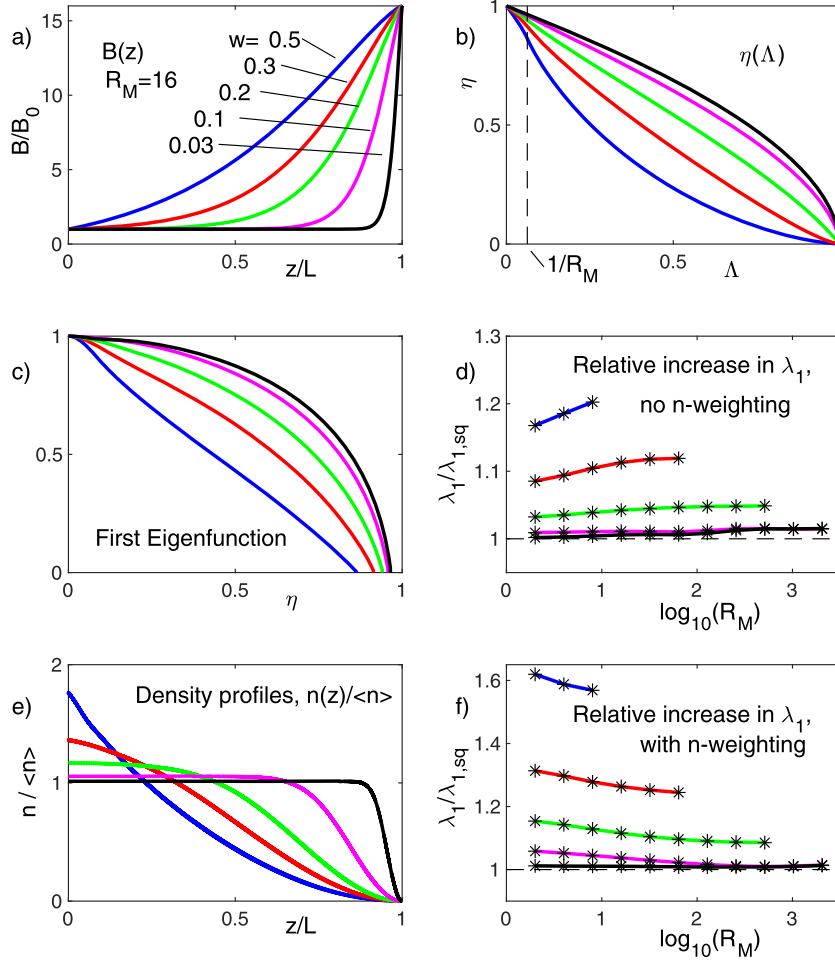


Figure 10. (a) Magnetic field profiles computed with equation (56) for $R_M = 16$ and $w \in \{0.03, 0.1, 0.2, 0.3, 0.5\}$. All panels apply the same color scheme as in (a) indicating the value of w . (b) Profiles of $\eta(\Lambda)$ calculated using equation (47) for the magnetic fields in (a). (c) First eigenfunctions $I_1(\eta)$ corresponding to the magnetic configurations in (a). (d) Ratios of $\lambda_1/\lambda_{1,\text{square}}$ as functions of $\log_{10}(R_M)$ for the considered values of w . (e) Using $f(v, \eta) \simeq f_1(v)I_1(\eta)$ the relative density profiles $n(z)/\langle n \rangle$ are computed and shown here. (f) Like (d), but here including the spatial variation in the scattering frequency imposed by $n(z)/\langle n \rangle$.

Let η_{TP} correspond to the trapped passing boundary also characterized by $\Lambda = 1/R_M$. We apply the fact that any arbitrary function defined for $0 \leq \eta \leq \eta_{\text{TP}}$ can be expanded in the set $M_m(\eta)$ such that

$$I_j(\eta) = \sum_m V_{jm} M_m(\eta). \quad (51)$$

Thus, for any particular value of j the coefficients in V_{jm} by equation (51) provide a representation of $I_j(\eta)$. Our aim is then to find the values of V_{jm} that makes equation (50) true.

Inserting equation (51) into equation (50) we obtain

$$\sum_m V_{jm} \langle \mathcal{L} \rangle_z M_m(\eta) = -\lambda_j \sum_n V_{jn} M_n(\eta). \quad (52)$$

Numerically, for each $M_m(\eta)$ we can compute $\langle \mathcal{L} \rangle_z M_m$ and expand the result in the $M_l(\eta)$ -basis functions such that

$$\langle \mathcal{L} \rangle_z M_m = \sum_l h_{ml} M_l, \quad (53)$$

where

$$h_{ml} = \frac{1}{\alpha_l} \int_0^{\eta_{\text{TP}}} (\langle \mathcal{L} \rangle_z M_m) M_l d\eta. \quad (54)$$

Inserting this into equation (52) we then get

$$\sum_m V_{jm} \sum_l h_{ml} M_l(\eta) = -\lambda_j \sum_n V_{jn} M_n(\eta).$$

Now, because M_l is an orthogonal set, this requires that

$$\sum_m h_{ml} V_{jm} = -\lambda_j V_{jl}, \quad (55)$$

and we have reduced the problem of finding the $I_j(\eta)$ eigenfunctions to simply solving a standard matrix eigenvalue problem.

4.5. Mirror confinement from non-uniform B

To explore numerically the impact on the mirror confinement of non-uniform magnetic fields we consider the following

analytical form for the magnetic field strength as a function of position z along the mirror with $-1 \leq z \leq 1$

$$B(z) = 1 + (R_M - 1) \frac{\exp\left(-\frac{\|z|-1\|^k}{w^k}\right) - \exp\left(-\frac{1}{w^k}\right)}{1 - \exp\left(-\frac{1}{w^k}\right)}. \quad (56)$$

Here, w and k are parameters that control the spatial profile of $B(z)$, with w representing the characteristic length scale for which the magnetic field is changing close to the mirror throat. For $R_M = 16$ and $k = 1.3$, profiles of $B(z)$ are shown in figure 10(a) for $w \in \{0.03, 0.1, 0.2, 0.3, 0.5\}$. Note that the square mirror is approached for $w \rightarrow 0$.

For each of the profiles in figure 10(a) we use equation (47) to evaluate $\eta(\Lambda)$ shown in figure 10(b). The values of $\eta_{TP} = \eta(1/R_M)$ describing the trapped/passing boundary are identified as the intersections of the curves with the vertical dashed line marked $1/R_M$. We observe how η_{TP} is different for each profile, decreasing with increasing w . This indicates that the confinement of the mirror decreases as the magnetic configurations depart more and more from the square magnetic geometry.

To quantify the effect of the departure from the square well geometry, we notice in equation (32) how Q_0 is inversely proportional to the first eigenvalue of the scattering operator, λ_1 . Thus, the impact of the non-square geometries can be assessed by examining λ_1 predicted for the orbit averaged scattering operator using equation (55). In figure 10(c) the curves represent the first eigenfunction, $I_1(\eta)$, for the magnetic profiles in figure 10(a). Meanwhile, in figure 10(d) the relative increase in the first eigenvalue, $\lambda_1/\lambda_{1,\text{square}}$ is shown as a function of $\log_{10}(R_M)$ again for $w \in \{0.03, 0.1, 0.2, 0.3, 0.5\}$. We observe that for $w \leq 0.2$ that $1 < \lambda_1/\lambda_{1,\text{square}} \leq 1.05$.

Configurations with $w \simeq 0.2$ are realistic to implement in the laboratory and the value of $\lambda_1/\lambda_{1,\text{square}} \simeq 1.05$ suggests that the impact on the mirror confinement is then relative insensitive to the departure from the square well. However, these estimates were computed with $n/\langle n \rangle = 1$, and do not include the changes in the density profile that go along with the non-square configuration. Assuming that $f(\eta, v) \simeq f_1(v)I_1(\eta)$, with the knowledge of the first eigenfunction, $I_1(\eta)$, we can estimate the relative density profiles, $n/\langle n \rangle$. For this we note that

$$d\frac{v_{\parallel}}{v} = \frac{1}{2\sqrt{1-B\Lambda/B_0}} \frac{B}{B_0} d\Lambda,$$

and using $\langle n \rangle = N/V$ together with equation (46) it follows that

$$\frac{n}{\langle n \rangle} = \langle 1 \rangle_{\Lambda} \frac{\int_{1/R_M}^{1/\tilde{B}} \frac{\tilde{B} I_1(\Lambda)}{\sqrt{1-\tilde{B}\Lambda}} d\Lambda}{\int_{1/R_M}^1 \tilde{\tau}_b I_1(\Lambda) d\Lambda}. \quad (57)$$

Profiles of $n/\langle n \rangle$ are shown in figure 10(e), again corresponding to the magnetic field configurations in figure 10(a).

Including this density weighting when computing the orbit averages as specified by equation (42), the value of $\lambda_1/\lambda_{1,\text{square}}$ increases as shown in figure 10(f). For example, the green curve corresponding to $w = 0.2$ now has a value of $\lambda_1/\lambda_{1,\text{square}} \simeq 1.15$, and the reduction in Q_0 is similar to that observed in the square geometry when reducing R_M by 30%.

For $w \in \{0.3, 0.5\}$ corresponding to the red and blue curves in figure 10(f), the values of $\lambda_1/\lambda_{1,\text{square}}$ are larger, and will significantly degrade the fusion performance of the device. The analysis therefore suggests that magnetic mirrors should be designed with profiles that resemble those of equation (56) for $w \leq 0.2$.

5. FBIS, a new fast beam ion solver

Compared to the reduced model in section 3, in this section we develop a more complete numerical model for beams of deuterium and tritium injected into a physical mirror geometry. We will denote this code as FBIS, standing for fast beam ion solver. The Fokker–Planck equation solved is similar to that considered in section 3, but here we include the effects of energy diffusion, a non-uniform magnetic field, and the electrostatic potential.

Initially, we neglect the effects of a distributed electrostatic potential, $\Phi(z)$, such that the solution for each of the species is separable and can be written on the form

$$f = \sum_j f_j(v) I_j(\eta). \quad (58)$$

Here I_j are the eigenfunctions of equation (50) with respect to the orbit averaged scattering operator. After this solution is obtained, an iterative scheme is provided by which the potential $\Phi(z)$ can be determined for the central mirror cell and continuously into the expander. In turn, the effect of the potential will also be approximately included in the beam ion distributions by updating the loss boundary in velocity space as appropriate.

5.1. Slow down on hot ions

In addition to considering the deuterium and tritium ions separately, we apply an enhanced representation of equation (19), generalized to included energy diffusion through the terms proportional to $\tilde{h}$ and $\tilde{g}_2$

$$\begin{aligned} & -\beta_m \lambda_j \frac{v_c^3 \tilde{g}_1}{v^3} f_j + \frac{1}{v^2} \frac{\partial}{\partial v} \left[(v^3 + v_c^3 \tilde{h}) f_j + \frac{v_c^3 \tilde{g}_2}{2} \frac{\partial f_j}{\partial v} \right] \\ & = -\tau_s S_j \frac{\delta(v - v_0)}{v^2}. \end{aligned} \quad (59)$$

Here

$$\tilde{h} = -v^2 \frac{\partial h}{\partial v}, \quad \tilde{g}_1 = \frac{\partial g}{\partial v}, \quad \tilde{g}_2 = v^2 \frac{\partial^2 g}{\partial v^2},$$

where we again apply the density normalized Rosenbluth potentials

$$\begin{aligned} g(v) &= \int f_1(v') |\mathbf{v} - \mathbf{v}'| d^3 v' / \int f_1(v') d^3 v', \\ h(v) &= \int \frac{f_1(v')}{|\mathbf{v} - \mathbf{v}'|} d^3 v' / \int f_1(v') d^3 v', \end{aligned}$$

and f_1 is the solution to equation (59) obtained for $\lambda_j = \lambda_1$.

For a given f_1 , the ODE in equation (59) is readily integrated numerically to obtain $f_j(v)$. We apply an iterative

scheme where equation (59) is first solved based on a guess for f_1 . The guess is then replaced by the new f_1 and so forth. A sufficiently converged solution is obtained after four iterations. Also, note that for an initial Maxwellian guess for f_1 we have

$$\tilde{h} = \Psi, \quad \tilde{g}_1 = v\Psi/x,$$

where the Maxwell integral is evaluated with $x = v^2/v_{ti}^2$

$$\Psi(x) \equiv \frac{2}{\sqrt{\pi}} \int_0^x dt \sqrt{t} e^{-t}. \quad (60)$$

To permit our application of the separation of variables technique, equation (59) does not include non-linear effects in the scattering rate related to the velocity space anisotropy of the beam-ion distributions. For the square-well configuration, however, this issue was investigated in reference [11], where the inclusion of the non-linear terms resulted in modest enhancements for the predicted values of Q_0 .

5.2. Numerical profiles illustrating the model

Before addressing the approach by which we determine the electrostatic potential, as well as the lost populations of electrons and ions, we find it useful to provide some numerical profiles and solutions. These results will then help aid the subsequent discussion of the remaining details of the model.

Figure 11(a) shows the magnetic field profile applied for the calculations in the present section, corresponding to $R_M = 16$ with more details given in the caption. The configuration is symmetric about $z = 0$ and we only show the part $z > 0$. The confining central part of the mirror is for $z/L \leq 1$, whereas $1 \leq z/L \leq 2$ represents the expanding exhaust field. At $z/L = 2$ we assume the presence of a wall absorbing all particles reaching it. To maintain quasi-neutrality and hold the electrons back so they leak out of the mirror at the same rate as the ions, the electrostatic potential in figure 11(b) develops. Later in section 5.6 we will describe the iterative scheme for determining $\Phi(z)$. In contrast to the more collisional GDT concept [7, 8] in the low-collisional mirror considered here, the main potential drop occurs inside the central cell and not in the expander. The total potential drop from $z = 0$ to the wall at $z/L = 2$ we denoted Φ_w and we observe $e\Phi_w \simeq -5T_e$, which is in agreement with predictions by [18].

As shown in figure 11(c), the ion density comprises the confined beam ions (red line), the beam ions scattered into the loss-cone (green line), and a thermal population trapped electrostatically by the slight non-monotonic potential profile (black line). For the present configuration the density of the thermal population is three orders of magnitude below the density of the confined beam ions and has no practical implications.

For $\Phi(z) = 0$ the solution of equations (58) and (59) is shown in figure 12(d), corresponding to a scenario with beam injection at 45° compared to the magnetic field. The peak in subplots (d) and (e) of f at $\Lambda \sim 0.5$ and $v/v_0 = 1$ is the region where particles are sourced. As will be detailed below, the electron temperature of $T_e/\mathcal{E}_{\text{beam}} = 0.15$ is determined by considerations similar to those in section 3. At this value of T_e ,

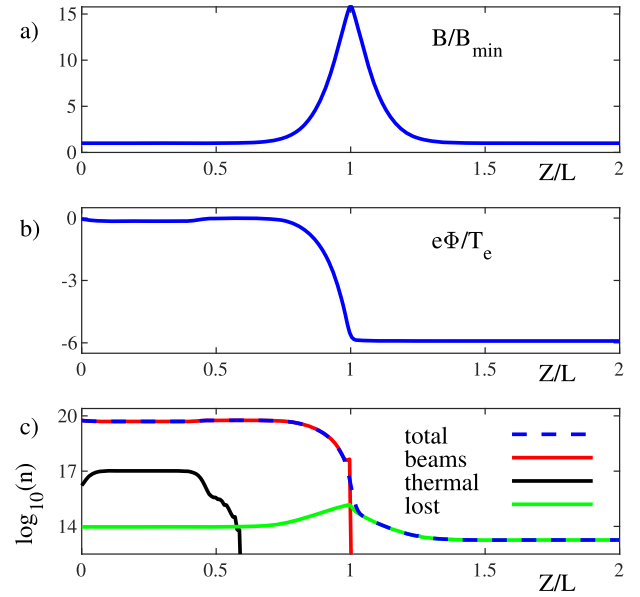


Figure 11. (a) Magnetic field profile corresponding to equation (56) with $R_M = 16$, $k = 1.3$ and $w = 0.1$. (b) Electrostatic potential $e\Phi/T_e$ inferred for maintaining quasi-neutrality of the plasma. (c) Profiles of $\log_{10}(n \text{ (m}^{-3}\text{)})$ for the various components contributing to the ion density. The profiles are obtained with the injection of 70 kV deuterium and tritium beams at 45° with the respect to the magnetic field. The total current of the beams is 1 A, into a magnetic mirror with a plasma volume of $\sim 0.075 \text{ m}^3$.

the effect of slow down on the electrons is clearly visible in the solution, which away from the source region is peaked at $\Lambda \simeq 1$ and $v/v_0 \simeq 0.65$. This dominant part of the distributions is dictated by the first eigenfunction $I_1(\eta)$ accompanied by a $f_1(v)$ similar to the curve in figure 6 for $\tilde{T}_e/\ln(R_M)^{2/3} = 0.05$. A main difference, however, is that due to energy diffusion f is now finite also for $v/v_0 > 1$.

With the knowledge of $f_0(v, \Lambda)$ the local distributions of confined ions can be computed at any location within the central cell of the device. Examples of $\log_{10}(f(v_{\parallel}, v_{\perp}))$ -contours are shown in figures 12(a)–(c) corresponding to locations with $\tilde{B} \in \{1.1, 4.5, 15.5\}$, respectively. The magenta dashed lines represent the trapped/passing boundaries and the regions with $v_{\perp}$ above these lines hold the confined ions. The figures also include lost ions with their values of f multiplied by a factor of 10^3 , such that they become clearly visible. For example, due to the enhancement by this factor, in figure 12(c) the lost ions to the right of the trapped/passing boundary is the main feature of the plot. We describe below how f is estimated within these loss regions.

5.3. Rate of ions lost across the losscone

In steady state, the rate at which ions are lost through the mirror throat and onto the expander is naturally equal to the beam current. However, in order to model the lost ions in the expander it is important to know their distribution as a function of energy. Again, we first evaluate the losses with the assumption that $\Phi(x) = 0$ within the center cell, and then later generalize these results to include the effects of $\Phi(x)$.

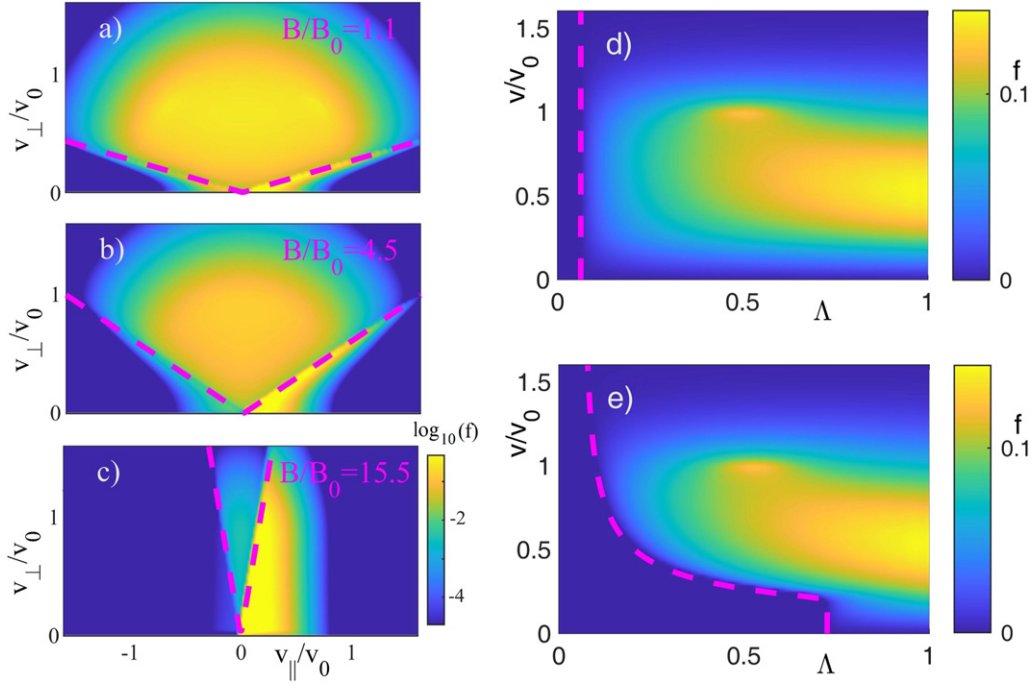


Figure 12. (a)–(c) Local beam ion distributions evaluated in the $(v_{\parallel}, v_{\perp})$ -plane. The magenta dashed lines mark the trapped/passing boundary for $\Phi = 0$. The phase space densities of the lost ions are multiplied up by a factor of 10^3 so they are more easily observed. (d) Solution of equation (58) for beam-injection at 45° relative to the magnetic field. (e) The solution in (d) updated with losses induced by $\Phi(z)$. The magenta dashed line is the trapped passing boundary approximately given by equation (71).

The local flux in velocity space due to pitch angle scattering is

$$\Gamma_{\xi} = \frac{\beta_m}{\tau_s} \frac{v_c^3 \tilde{g}_1}{v^2} \frac{\partial f}{\partial \theta} \equiv \tilde{\Gamma}_{\xi} \frac{\partial f}{\partial \theta}.$$

Here $\theta = \arccos(v_{\parallel}/v)$ is the pitch angle, and with $\tilde{B} = B/B_{\min}$ and using $\mathcal{E}$ and Λ as the principle variables it follows that

$$\frac{\partial}{\partial \theta} = \frac{2\sqrt{\tilde{B}\Lambda}\sqrt{1-\tilde{B}\Lambda}}{\tilde{B}} \frac{\partial}{\partial \Lambda}.$$

Then, considering a volume element d^3x , to compute the number of particles per unit time lost from this volume out of the mirror, we need to integrate Γ_{ξ} over the loss cone, $\Lambda_M = 1/R_M$, in velocity space

$$\begin{aligned} \frac{dN_{\text{loss},i}}{dt d^3x} &= \Gamma_{\xi} 2\pi v \sin \theta dv|_{\Lambda_M}, \\ &= 4\pi \tilde{\Gamma}_{\xi} v dv \Lambda \sqrt{1-\tilde{B}\Lambda} \frac{\partial}{\partial \Lambda} f \Big|_{\Lambda_M}. \end{aligned}$$

Using that $d^3x = A dz = (A_0 B_0/B) dz$, the total loss within a differential velocity bin dv becomes

$$\begin{aligned} \frac{dN_{\text{loss},i}}{dt dv} &= 4\pi A_0 \tilde{\Gamma}_{\xi} v \frac{\partial}{\partial \Lambda} f \Big|_{\Lambda_M} \int dz \frac{B_0^2}{BB_M} \sqrt{1-\frac{B}{B_M}}, \\ &= 4\pi A_0 \tilde{\Gamma}_{\xi} l \frac{B_0}{B_M} G(l) v \frac{\partial}{\partial \Lambda} f \Big|_{\Lambda_M}, \end{aligned} \quad (61)$$

where

$$G(z) = \frac{1}{l} \int_{-l}^z dz' \frac{B_0}{B(z')} \sqrt{1-\frac{B(z')}{B_M}}. \quad (62)$$

5.4. Model for lost ions

Above we determined $dN_{\text{loss},i}/(dt dv)$ as a function of energy, $\mathcal{E}$. We assume that these lost ions reach the mirror throat with total energy $U = \mathcal{E} + e\Phi$, and a parallel temperature $T_L \ll \mathcal{E}_{\text{beam}}$. The lost ion distribution may then be expressed by a function of the form

$$f_L^+(U, \Lambda, z) = H(U) \exp\left(-\frac{U(1-\Lambda B_M/B_0)}{T_L}\right) G(z), \quad (63)$$

applicable for values of Λ in the loss cone, $\Lambda \leq \Lambda_M$. The function $H(U)$ is determined by matching the fluxes of f_L^+ to the fluxes of lost ions at the throat of the mirror. First, the parallel particle flux of the form in equation (63) is readily evaluated:

$$\begin{aligned} \frac{dN_{f_L^+}}{dt dv_{\perp}} &= A_0 \frac{B_0}{B_M} \int_0^{\infty} v_{\parallel} f_L^+ 2\pi v_{\perp} dv_{\parallel}, \\ &= A_0 \frac{B_0}{B_M} H G(l) v_{\perp} 2\pi \int_0^{\infty} v_{\parallel} \exp\left(-\frac{v_{\parallel}^2}{v_{Lt}^2}\right) dv_{\parallel}, \\ &= A_0 \frac{B_0}{B_M} H G(l) v_{\perp} \pi v_{Lt}^2. \end{aligned}$$

Matching this with the fluxes in equation (61) we find

$$A_0 \frac{B_0}{B_M} H G(l) \pi v_{\perp} v_{Lt}^2 = 4\pi A_0 \tilde{\Gamma}_{\xi} v \frac{\partial}{\partial \Lambda} f \Big|_{\Lambda_M} l \frac{B_0}{B_M} G(l)$$

such that

$$\begin{aligned} H(U) &= 4\tilde{\Gamma}_\xi \frac{v}{v_\perp v_L^2} l \frac{\partial}{\partial \Lambda} f \Big|_{\Lambda_M}, \\ &= 4 \frac{\beta_m v_c^3}{\tau_s v^2 v_L^2} l \frac{\partial}{\partial \Lambda} f \Big|_{\Lambda_M}. \end{aligned} \quad (64)$$

Note that $H(U)$ is evaluated as a function of $U = \mathcal{E} + e\Phi$. Meanwhile, the majority of the ions lost in the mirror scatter across the loss cone where the density is large and $e\Phi \ll T_i$. Thus, it is therefore appropriate to evaluate the rhs of equation (64) with $\mathcal{E} = U$. In contrast, with $H(U)$ determined, equation (63) should then be evaluated with $U = \mathcal{E} + e\Phi$, such that the energy boost ions observe as they approach the mirror throat and continue into the expander is accurately included. Again, in section 5.6 below we will specify how to determine the profile of $\Phi(z)$. For $\Phi = 0$, examples of f_L^+ were included in figures 12(a)–(c), while the profiles (to be discussed later) in figure 13 include the effects of $\Phi(x)$.

5.5. Lost electrons

We next determine the electron fluxes into the loss cone with $m_e v^2/2 > e\Phi_w$. It is convenient to introduce $v_w = \sqrt{2e\Phi_w/m_e}$, such that this condition simply becomes $v > v_w$. Similar to the case for the ions, we consider again a volume element d^3x , to compute the number of particles per unit time lost from this volume out of the mirror, we need to integrate Γ_ξ over the loss cone, $\Lambda_M = 1/R_M$, in velocity space

$$\begin{aligned} \frac{dN_{\text{loss},e}}{dt d^3x} &= \Gamma_\xi 2\pi v \sin \theta dv \Big|_{\Lambda_M}, \\ &= 4\pi \tilde{\Gamma}_\xi v \Lambda \sqrt{1 - \tilde{B}\Lambda} \frac{\partial}{\partial \Lambda} f \Big|_{\Lambda_M}. \end{aligned}$$

Again, with $d^3x = A dz = A_0 B_0/B dz$, the total losses become

$$\begin{aligned} \frac{dN_{\text{loss},e}}{dt dv} &= 4\pi A_0 \tilde{\Gamma}_\xi v^2 \int_{\tilde{B}\Lambda \leq 1} dz \frac{\Lambda}{B} \sqrt{1 - \tilde{B}\Lambda} \frac{\partial}{\partial \Lambda} f \Big|_{\Lambda_M}, \\ &= 4\pi A_0 \tilde{\Gamma}_\xi v^2 \frac{\partial}{\partial \Lambda} f \Big|_{\Lambda_M} \int dz \frac{B_0^2}{BB_M} \sqrt{1 - \frac{B}{B_M}}, \\ &= 4\pi A_0 \tilde{\Gamma}_\xi v^2 \frac{\partial}{\partial \Lambda} f \Big|_{\Lambda_M} l \frac{B_0}{B_M} G(l), \end{aligned} \quad (65)$$

where $G(z)$ is given in equation (62).

For the electrons $\tilde{\Gamma}_\xi = 0.5(\nu_\perp^{e/e} + \nu_\perp^{e/i})v$, where $\nu_\perp^{e/e} \simeq \nu_\perp^{e/i} \simeq 2\tilde{\nu}_0^{ee} v_i^3/v^3$ with

$$\tilde{\nu}_0^{ee} = \frac{ne^4 \log \Lambda}{4\pi \varepsilon_0^2 m_e^2 v_i^3} \left(= \frac{1}{\tau_s} \frac{3\sqrt{\pi} m_f}{4z_f^2 m_e} \right). \quad (66)$$

For energies larger than $e\Phi_w$ we assume that the confined electrons fulfill $f(v, \Lambda) = f(v)I_1(\Lambda)$, such that

$$\frac{\partial}{\partial \Lambda} f \Big|_{\Lambda_M} = f \frac{\partial I_1}{\partial \Lambda} \Big|_{\Lambda_M}.$$

Equation (65) then becomes

$$\begin{aligned} \frac{dN_{\text{loss},e}}{dt dv} &= 8\pi A_0 \tilde{\nu}_0^{ee} \frac{v_i^3}{v} f \frac{\partial I_1}{\partial \Lambda} \Big|_{\Lambda_M} l \frac{B_0}{B_M} G(l) \\ &= H_e \frac{v_i}{v} f, \end{aligned} \quad (67)$$

with

$$H_e = 8\pi A_0 \tilde{\nu}_0^{ee} v_i^2 \frac{\partial I_1}{\partial \Lambda} \Big|_{\Lambda_M} l \frac{B_0}{B_M} G(l).$$

The total flux of electrons is then

$$\frac{dN_{\text{loss},e}}{dt} = \int_{v_w}^{\infty} \frac{dN_{\text{loss},e}}{dt dv} dv.$$

For a Maxwellian $f(v)$ with density n_0 this expression involves an integral of the form

$$\int_a^{\infty} \frac{1}{x} \exp(-x^2) dx \simeq \exp(-a^2) \left(\frac{1}{2a^2} - \frac{1}{4a^4} \right).$$

Retaining only the first term we find

$$\begin{aligned} \frac{dN_{\text{loss},e}}{dt} &\simeq H_e \frac{n_0}{\sqrt{\pi}^3} \frac{\exp(-v_w^2/v_i^2)}{2v_w^2}, \\ &= \frac{4n_0 A_0}{\sqrt{\pi}} \tilde{\nu}_0^{ee} \frac{\partial I_1}{\partial \Lambda} \Big|_{\Lambda_M} l \frac{B_0}{B_M} G(l) \frac{v_i^2}{v_w^2} \exp\left(-\frac{v_w^2}{v_i^2}\right). \end{aligned} \quad (68)$$

This loss current must match the beam ion current, $e dN_{\text{loss},e}/dt = I_{\text{beam}}$, and can be applied for determining $e\Phi_w$ as a function of T_e .

Likewise, the energy loss is

$$P_{\text{loss},e} = \int_{v_\phi}^{\infty} \frac{dN_{\text{loss},e}}{dt dv} \frac{m_e(v^2 - v_w^2)}{2} dv$$

and this expression, thus, involves an integral of the form

$$\int_a^{\infty} \frac{x^2 - a^2}{x} \exp(-x^2) dx \simeq \exp(-a^2) \left(\frac{1}{2a^2} - \frac{1}{4a^4} \right).$$

Again, retaining only the first term, it then follows that

$$\frac{P_{\text{loss},e}}{dN_{\text{loss},e}/dt} \simeq \frac{0.5m_e v_i^2 \exp(-a^2)(1/(2a^2))}{\exp(-a^2)(1/(2a^2))} = T_e$$

such that

$$P_{\text{loss},e} \simeq I_{\text{beam}} T_e / e. \quad (69)$$

This is the electron loss at the end wall, while the ion power at the end wall includes also the full potential difference ($U = \mathcal{E} + \Phi$, see section 5.4). The energy losses must be supplied by their beam heating. Similar to section 3, this then provides a constraint by which we determine T_e .

5.6. Iterative scheme for finding $\Phi(z)$

The electrostatic potential develops and adjusts to maintain quasi-neutrality between the electrons and ions. We use $\tilde{v} = v/v_{te}$ and similar to above $\tilde{v}_w = \sqrt{2e|\Phi_w|/(m_e T_e)}$ denotes

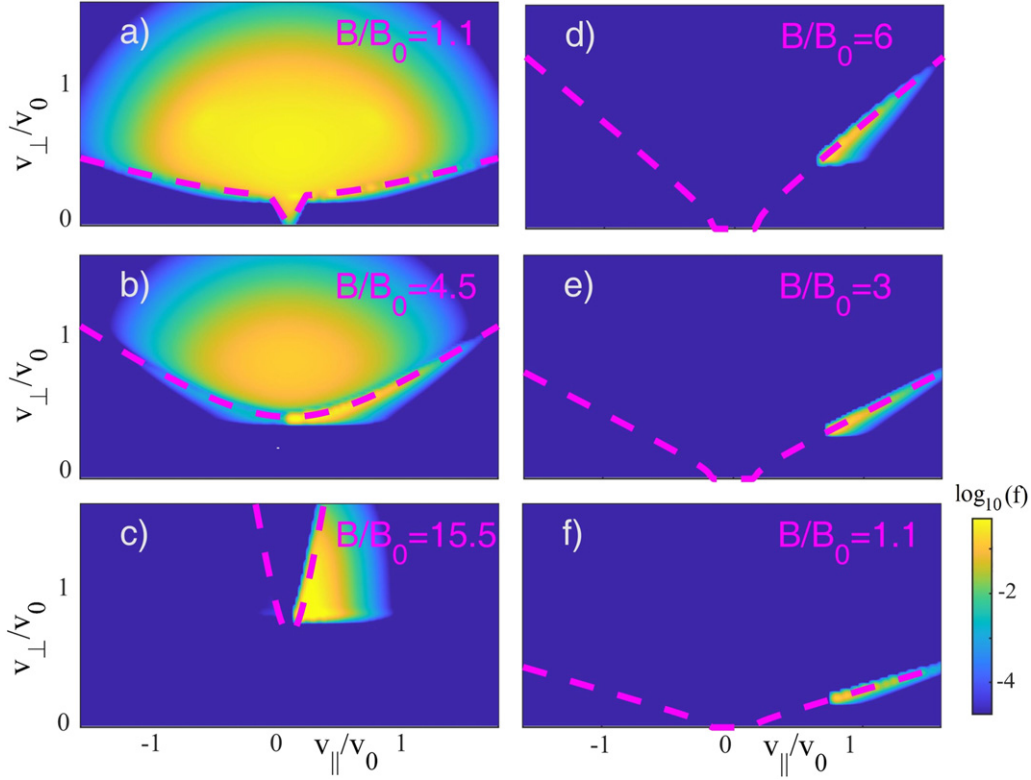


Figure 13. (a)–(c) Local beam ion distributions inside the central mirror cell. (d)–(f) Local beam ion distributions in the mirror exhaust. In all panels, the magenta lines are the trapped passing boundaries including the effects of $\Phi(z)$. To make their features visible, the lost ion distributions are multiplied up by a factor of 10^3 .

the characteristic normalized speed of Φ_w , and for positions along the length of the mirror $\tilde{v}_\phi(z) = \sqrt{2e|\Phi(z)|/(m_e T_e)}$. In addition, $\tilde{v}_\Delta = \sqrt{\tilde{v}_w^2 - \tilde{v}_\phi^2}$ is the normalized speed corresponding to the energy difference $|e\Phi_w| - |e\Phi|$. We assume that the electrons confined by Φ_w are Maxwellian distributed, such that the electron density fulfills

$$\begin{aligned} n_e(z) &= \frac{n_0}{\pi^{3/2}} \int_0^{\tilde{v}_\Delta} 4\pi \tilde{v}^2 \exp(-(\tilde{v}_\phi^2 + \tilde{v}^2)) d\tilde{v}, \\ &= n_0 \exp(-\tilde{v}_\phi^2) [\text{erf}(\tilde{v}_\Delta) - 2\tilde{v}_\Delta \exp(-\tilde{v}_\Delta^2)]. \end{aligned} \quad (70)$$

With equation (70) for a given profile of $\Phi(z)$ the electron density profile is now readily evaluated. The next task is then to find numerical methods to determine the ion density also as a function of $\Phi(z)$, and then search for the profile of $\Phi(z)$ that ensure $n_e(z) = n_i(z)$ along the full length of the mirror.

With the knowledge from section 3 that $T_{i,\text{eff}} \gg T_e$, the ion density is not expected to be significantly affected by $\Phi(z)$. To approximately account for the changes in $f_i(U, \Lambda)$ caused by the $\Phi(z)$ we notice that the trapped passing boundary, Λ_{TP} changes due to $\Phi(z)$. The form of Λ_{TP} can be found by analyzing the spatial structure of the effective parallel potential $e\Phi + \mu B$ for each value of μ [22]. For most energies, however, the following expression is accurate

$$\Lambda_{\text{TP}} = \frac{1}{R_M} \left(1 - \frac{e\Phi_{\text{throat}}}{U} \right), \quad (71)$$

where again $U = \mathcal{E} + e\Phi$. This form of Λ_{TP} approximately coincides with the magenta dashed line in figure 12(f), except the deconfining potential only becomes significant for $B \gtrsim 3/2$, and all ions with $\Lambda \gtrsim 2/3$ remain confined (this explains the vertical section of the magenta dashed line in figure 12(e)).

Let $f_i^\Phi(U, \Lambda)$ denote the distribution including the effects of $\Phi(z)$. As a somewhat crude approximation we then use $f_i^\Phi(U, \Lambda) = f_i(\mathcal{E}, \Lambda^*)$, where f_i is the solution in equation (58) evaluated with $\mathcal{E} = U$ and

$$\Lambda^* = 1 - \frac{1 - 1/R_M}{1 - \Lambda_{\text{TP}}} (1 - \Lambda). \quad (72)$$

As is evident in figure 12(f), for a given $\mathcal{E}$ the effect of using Λ^* is to compress the previous solution into the reduced range $\Lambda_{\text{TP}} \leq \Lambda \leq 1$ that remains magnetically confined despite the deconfining electrical forces.

The enhanced loss-rate caused by the increased loss-regions in velocity space can be accounted for when solving equation (59), simply by letting Λ_1 be a function of energy and increase λ_1 for the most affected lower energies $\mathcal{E} < |e\Phi_{\text{throat}}|/R_M$. For example, for $R_M = 16$ we found it appropriate to increase λ_1 in this energy range to the value of λ_1 observed at $R_M = 1.5$. This was motivated by the $\Phi(z)$ -profile in figure 11(b), where $|\Phi|$ only becomes significant for $B/B_0 > 1.5$. In any case, adjustments of this kind are found to have negligible impact on our numerical solutions as well as the estimated fusion performance.

Using the above method for obtaining $f_i^\Phi(U, \Lambda)$ for a specified $\Phi(z)$ -profile, we may now also evaluate the local distributions along the mirror. Examples are shown in figure 13, where in figures 13(a)–(c) we again consider positions inside the central cell with $B \in \{1.1, 4.5, 15.5\}$. Meanwhile, figures 13(d)–(f) are positions in the exhaust with $B \in \{6, 3, 1.1\}$, documenting the evolution of the lost ion distribution (which again are multiplied up by a factor of 10^3).

Along the mirror we evaluate 100 distributions similar to those in figure 13, which in turn allow us to compute $n_i(z)$ corresponding to any particular profile of $\Phi(z)$. Using equation (70) we may then solve for an improved new profile of $\Phi(z)$. This scheme for determining the profile of $\Phi(z)$ consistent with quasi-neutrality is found to be well converged within 12 such iterations.

As the reader may have noticed, the profile of n_e in equation (70) is only determined to within the scalar factor of n_0 . Likewise, τ_s within equation (59) also depends on n_0 . In fact, consistent with the reduced model in section 3 the total ion density scales as $N_{i,\text{tot}} \propto 1/N_{e,\text{tot}}$, while the shape of the relative profiles are insensitive to $N_{e,\text{tot}}$. Therefore, we may solve equation (59) with a random guess for $\langle n_e \rangle = N_{e,\text{tot}}/V$, and obtain the corresponding value for $N_{i,\text{tot}}$. Hereafter, we can re-scale the solution of equation (59) (without computational cost) to ensure that $N_{i,\text{tot}} = N_{e,\text{tot}}$. This step is also included in each cycle of the iterative process above for determining $\Phi(z)$.

5.7. Φ_w and the Pastukhov factor

In section 5.5 we described how the potential drop Φ_w from the center of the device to the material wall at the end of the expander is determined by using equation (68) and requiring that $e dN_{\text{loss},e}/dt = I_{\text{beam}}$. It should be noted that this constraint does not depend on the details of the $\Phi(z)$ -profile and can be solved independently. In fact, Φ_w is needed in equation (70), which is a key equation for obtaining $\Phi(z)$.

Another approach for estimating Φ_w is to require that

$$\frac{dN_{\text{loss},e}}{dt} = \int \frac{dN_{\text{loss},i}}{dt} dv.$$

Using equations (61) and (68) this condition can be rearranged to give

$$\frac{e\Phi_w}{T_e} \exp\left(\frac{e\Phi_w}{T_e}\right) = K \frac{\tau_{ii}}{\tau_{ee}}, \quad (73)$$

where

$$K = \frac{\langle n \rangle \frac{\partial I_1}{\partial \Lambda} \Big|_{\Lambda_M}}{\pi^{3/2} v_{ii}^3 \int dv \frac{\partial I_1}{\partial \Lambda} f \Big|_{\Lambda_M}}. \quad (74)$$

Generally, we find that $K \simeq 1$ and equation (73) then reproduces the Pastukhov factor that originally was calculated for the square-well mirror. In a practical sense, the potential required to confine the electrons can approximately be determined from the Pastukhov factor [18, 23, 24].

5.8. Fusion power and the self-consistent value of T_e

The total fusion yield of a particular configuration can be evaluated directly from the local distributions of the type

in figure 13. From the separate solutions for deuterium and tritium ions we apply the σ_{DT} fusion cross-sections provided by [25] and carry out the six-fold velocity space integral over the distributions to obtain the local fusion rate. These local rates are evaluated at 100 points axially along the device and integrated to yield the total fusion power, P_{fusion} .

Through τ_s and v_c , the electron temperature is in principle a free parameter when solving equation (59). However, with the knowledge of P_{fusion} we implement a simple numerical search to find the value of T_e consistent with the power-balance constraint in equation (24). The full numerical model is implemented in Matlab. While this implementation is not optimized for computational efficiency, fully converged solutions (including the search for the self-consistent T_e) are typically obtained in 3 min on a laptop computer.

6. Fusion gain of the magnetic mirror

In this final section we will apply the new numerical model presented above to evaluate a few important plasma parameters, including the fusion gain Q_0 , that can be expected for a magnetic mirror geometry realizable in the laboratory using modern superconducting coil technology. We apply magnetic field configurations corresponding to the $B(z)$ in equation (56) evaluated with $k = 1.3, w = 0.1$ and $R_M \in \{4, 8, 16\}$. For example, the profile with $R_M = 16$ is shown by the magenta line in figure 10.

In figure 14 the profiles presented are independent of the plasma volume and beam current, and are all evaluated with the beam injected at 90° compared to the direction of the magnetic field, yielding the best fusion performance. We emphasize that the study represents the absolute best case scenario where the mirror end-losses included in the model dominate all other loss channels. Of course, for an actual experiment this may not be the case. Thus, the profiles represent the upper limit set by classical transport on the magnetic fusion potential for the magnetic mirrors, which can be degraded by MHD or kinetic instabilities, fast ion non-adiabaticity, as well as other radial transport mechanisms. Preliminary estimates in reference [4] are promising.

In general we find that the reduced model in section 3 is in good agreement with the profiles in figure 14. For example, consider the curve in figure 14(a) for $R_m = 16$. This curve can be compared to a vertical cut at $\log_{10}(R_M) \simeq 1.2$ in figure 9(c), where the maximum value of $Q_0 \simeq 1.8$ is observed at $\mathcal{E}_{\text{beam}} \simeq 300$ keV. The profiles of Q_0 are also consistent with earlier results in reference [11].

The ion temperature in figure 14(b) is on the order of $T_i \simeq 0.6\mathcal{E}_{\text{beam}}$, which because of energy diffusion included in the numerical model is larger than the $\tilde{T}_i \simeq 0.4$ predicted by the reduced model. Meanwhile, the electron temperature in figure 14(b) is on the order of $T_e \simeq 0.14\mathcal{E}_{\text{beam}}$ and is consistent with the reduced model. The wall potential $\Phi_w \simeq 5.2T_e/e$ in figure 14(c) is again consistent with the results from the reference [18]. Likewise, the inferred values of the confinement time coefficient $\mathcal{C}$ of equation (31) fall within the range previously determined [12].

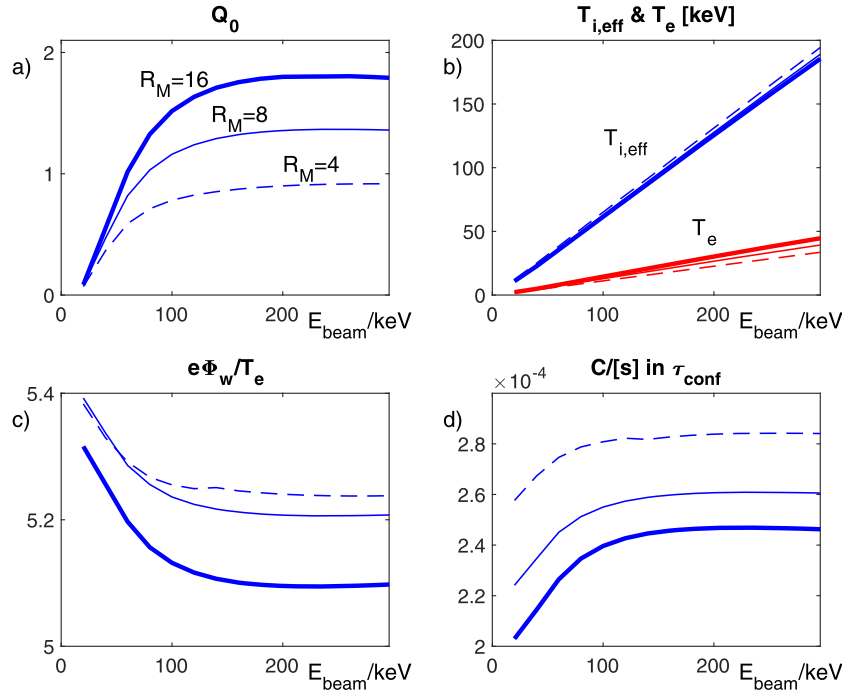


Figure 14. Predicted profiles as a function of the beam energy, $\mathcal{E}_{\text{beam}}$, for magnetic configurations corresponding to $R_M \in \{4, 8, 16\}$. The beam ions are injected at 90° with respect to the magnetic field. Panel (a) shows the fusion gain $Q_0 = P_{\text{fusion}}/P_{\text{beam}}$, and (b) displays both $T_{i,\text{eff}}$ and T_e . In panel (c) we show the normalized wall potential Φ_w/T_e , which develops to confine the electrons. Panel (d) provides the profiles of the coefficient C in the scaling law of the ion confinement time (i.e. equation (31)), observed to be a weak function of R_M .

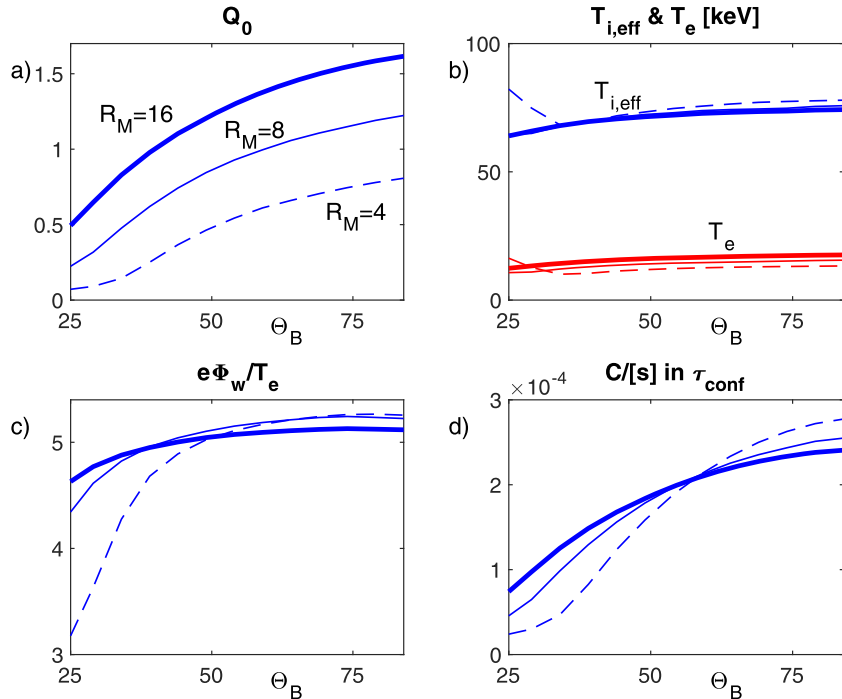


Figure 15. Similar to figure 14, but here as a function of the beam injection angle Θ_B for $\mathcal{E}_{\text{beam}} = 120$ keV.

The performance of the magnetic mirrors are sensitive to the beam ion injection angle, Θ_B . In figure 15 the same quantities as in figure 14 are evaluated but now as a function of Θ_B for $\mathcal{E}_{\text{beam}} = 120$ keV. The best performance is observed at perpendicular injection, $\Theta_B = 90^\circ$, and the value of Q_0 and C

declines rapidly as Θ_B approaches angles closer to the losscone in velocity space. Numerically, this effect is captured when calculating the source expansion factors S_j in equation (11). Also, the reduced model will capture the effects with the simple generalization of replacing S_1 in equation (12) with

the first term of equation (11). From that is clear that for the case where the source is peaked at the angle Θ_B , the performance of the mirror will scale roughly as $I_1(\cos(\Theta_B))$, where $I_1(\eta)$ is the first eigenfunction of the orbit averaged collision operator.

We emphasize again that the above numerical results do not represent a prediction for a particular mirror experiment. Rather, the aim of the analysis is to evaluate the performance of the mirror under the very best theoretical circumstances. In addition to plasma instabilities other effects may also degrade the viability and confinement of the magnetic mirror. For example, throughout our analysis we have assumed a perfectly pure DT plasma with $z_{\text{eff}} = \sum_j n_j z_j^2 / n_e = 1$. Because the potential $\Phi \simeq -5T_e/e$ confining the electrons will help eject impurity ions values of $z_{\text{eff}} \simeq 1$ are expected. Nevertheless, we may evaluate the effect of z_{eff} by noticing that through β_m in equation (1) the effective scattering rate is enhanced linearly with z_{eff} , and can be accounted for by replacing in all equations the eigenvalues λ_j by $z_{\text{eff}} \lambda_j$. In turn, this is equivalent to replacing $\log_{10}(R_M)$ by $\log_{10}(R_M)/z_{\text{eff}}$. A magnetic mirror with $R_M = 100$ and $z_{\text{eff}} = 2$ will then have a predicted performance reduced to that of a mirror with a pure DT plasma at $R_M = 10$.

As a useful property of the magnetic mirror, the theoretical values of Q_0 are independent of the plasma volume and the current in the neutral beams. However, for a reactor other physical constraints will set the minimum device size. For example, at a beam energy on the order of $\mathcal{E}_{\text{beam}} = 100$ keV to reduce the shine-through fraction to levels less than, say, 5%, the line-integrated plasma density along the path of the beams must be on the order of $\int n d\ell_{\text{beam}} \simeq 1.5 \times 10^{20} \text{ m}^{-2}$. Larger shine-through fractions will not only reduce Q_0 , but may also represent a significant problem to the integrity of the vacuum vessel components.

7. Conceptual design point for the WHAM++ experiment

The FBIS code is presently being used to design a breakeven experiment, WHAM++, as a follow on the prototype experiment WHAM under construction at UW-Madison. The tentative parameters include a central cell magnetic field $B_0 = 2.5$ T and high temperature superconducting mirror coils that are just becoming technically feasible to provide mirror fields of $B_M = 25$ T: the mirror ratio will be on the order of $R_M = 10$ which will increase with plasma β as $R_M \propto 1/\sqrt{1-\beta}$. The device is conceived to be powered mainly by NBJ at $\mathcal{E}_{\text{beam}} = 120$ keV and rely on perpendicular injection to optimize the fusion yield. The numerical results suggest that the electrons are electrostatically confined and their losses are limited to $5kT_e/\text{lost ion}$, while the fast ion distribution are likely to be dominated by pitch angle scattering rather than a sloshing distribution governed by electron drag. We note that this may be the first mirror experiment to operate in this regime.

The size of the plasma will be $a > 0.3$ m to be sufficiently large to attenuate the high energy neutral beam, but otherwise as compact as possible to minimize cost. The half length of the device currently has a design value of $L = 2.5$ m to stabilize

$m > 2$ modes through FLR effects by satisfying the stability condition

$$m^2 > \frac{a^2}{L\rho_i} = \frac{a/\rho_i}{L/a}. \quad (75)$$

The $m = 1$ stability will be addressed in a subsequent publication and be derived from a combination of vortex stabilization, active feedback [26], non-paraxial effects including use of divertors, and shape optimization.

Kinetic stability requires minimizing the ion ambipolar hole to stabilize the DCLC instability or by making the plasma sufficiently large. Sloshing ions provide an off axis potential peak that can electrostatically confine a warm ion species as has been demonstrated in 2XIIIB and TMX-U endplugs, and quite likely in the GDT as well that has empirically demonstrated this effect. We note that high mirror ratios considered here also likely help to reduce the ambipolar hole as seen in equation (71).

The plasma volume is then $V = \pi a^2 2L \simeq 1.4 \text{ m}^3$. The beam current, I_{beam} , will be adjusted to reach a normalized plasma pressure corresponding to $\beta \simeq 0.5$. We here apply the results from our numerical model to estimate the required value of I_{beam} . At the specified geometry the model predicts the absolute temperatures $T_{i,\text{eff}} = 75.4$ keV and $T_e = 16.3$ keV. With these the plasma density yielding the desired normalized pressure is readily computed $n = \beta(B_0^2/(2\mu_0(T_e + T_{i,\text{eff}}))) = 8.9 \times 10^{19} \text{ m}^{-3}$, which is sufficient to attenuate the NBI by $\sim 80\%$ over 0.6 m.

From the model an estimate is also available for the confinement time using equation (31) with the coefficient $\mathcal{C} = 2.5 \times 10^{-4}$ s. The required beam current can be found from $N = nV = \mathcal{C}(I_{\text{beam}}/e)(\mathcal{E}_{\text{beam}}/1 \text{ keV})^{3/2} \log_{10}(R_M)/n_{20}$ or

$$I_{\text{beam}} = \frac{Venn_{20}}{\mathcal{C} \log_{10}(R_M)(\mathcal{E}_{\text{beam}}/1 \text{ keV})^{3/2}} \simeq 53.4 \text{ A}.$$

Thus, similar to the NSTX experiment [27], the WHAM++ device will tentatively be powered by a mixture of deuterium and tritium beams at $\mathcal{E}_{\text{beam}} = 120$ keV with a total current of $I_{\text{beam}} = 50$ A. Under the outlined best case scenario, the fusion gain will be on the order of the injected beam, ($Q_0 \simeq 1$). The spatial distribution of plasma pressure and ion distributions are expected to be similar to those presented in figures 11 and 13.

8. Discussion and conclusions

This paper aspires to gain the attention of mirror specialists and the wider fusion community of a ‘revolution’ in mirror physics initiated by the new high field superconductors and the achievement of 1 keV electrons in the small GDT mirror facility at Novosibirsk. Specifically, with high field mirror/cusps, known techniques for stabilizing the ion cyclotron instabilities that limited performance of mirrors in the 1970s–80s can be implemented [4]. Given stability, mirror design concerns only collisional end losses using the Fokker–Planck equation. This paper describes in detail a Fokker–Planck code FBIS that will be calibrated on the new high field mirror experiment WHAM at the University of Wisconsin-Madison.

The classic problem of computing fusion gain for fast ions injected into a simple mirror has been revisited, starting with a simple analytic model and then generalizing to the physical geometry of a real mirror magnetic field with bounce-averaged collision operators appropriate for the weakly collisional mirror. We have considered a range of mirror ratios that may now be possible because of technological advances in superconducting materials and physics advances that have shown axisymmetric mirrors can be MHD and kinetically stable. A high mirror ratio and high field simple mirror likely represents the lowest capital cost approach to a breakeven experiment in part because the conditions for $Q_0 > 1$ are theoretically independent of density, power level, and machine size. This provides a modest opportunity for near term demonstration of both energy breakeven, but also for implementation of an efficient neutron source that could be utilized for blanket testing and technology demonstrations needed to close the fusion fuel cycle.

The code and mathematical framework developed, FBIS, also provide a starting point for a series of calculations and results to follow in subsequent publications. The solution can first be used to compute $P_{\perp}(\psi, B)$ and $P_{\parallel}(\psi, B)$ and be used iteratively to compute high beta equilibrium associated with the anisotropic pressure profile. This in turn should provide a starting point for global MHD stability analysis for low and intermediate m in a non-paraxial, high- β equilibria that can be used to optimize the plasma shape. In addition, local stability of the distribution function could be evaluated. Finally, FBIS and future modifications can be used for mirror reactor calculations including alpha heating at beta-enhanced mirror ratios. Preliminary estimates suggest effective Q_0 values of order $Q_0 \gg 1$, yielding low-cost DT mirror reactors [4]; and finally tandem mirrors burning DD or other advanced fuels [28].

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